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Engineering NIR-Sighted Bacteria

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This study establishes bathy phytochromes, a unique class of bacterial photoreceptors that respond to near-infrared light (NIR), as **important** tools for bacterial optogenetics. NIR light is a key control signal in optogenetics due to its deep tissue penetration and the ability to combine with existing red- and blue-light sensitive systems, but thus far, NIR-activated proteins have been poorly characterized. The strength of the evidence is **solid** overall, with comprehensive in vitro characterization, modular design strategies, and validation across different hosts. There are some questions that remain – such as the rationale for linker choices, characterization of growth and performance relative to controls, and the physiological significance of color blind effects at alkaline pH – but overall, this study should advance the fields of optogenetics and photobiology and inspire future work.

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Abstract

Spatially and temporally orchestrated gene expression underpins organismal development, physiology, and adaptation. In bacteria, two-component systems (TCS) translate environmental cues into inducible expression outputs. Inducible expression also serves as a versatile implement in both basic and applied science. Here, we harness the photosensors of rhizobial bathy-phytochromes to construct synthetic TCSs for stringent activation of gene expression by near-infrared (NIR) light in laboratory and probiotic *Escherichia coli* strains, and in *Agrobacterium tumefaciens*. Orthogonal TCSs afford the multiplexed expression control of several genes by NIR and visible light. Notwithstanding substantial photochemical activation of bathy-phytochromes by visible radiation, the NIR-light-responsive systems hardly responded to red light. Evidently, light signals can be processed by TCSs into highly nonlinear responses at the physiological relevant level of gene expression. These fundamental aspects likely extend to naturally occurring TCSs. Depending on their photosensor traits and environmental conditions, bathy-phytochromes may thus either be NIR-specific or function as colorblind receptors of light vs. darkness.

Introduction

Proportionate to its essential roles in organismal physiology, development, replication, and adaptation, gene expression is stringently and precisely regulated. Various cellular circuits integrate cognate input signals and, in turn, ramp up or down the expression of target genes. Molecular analyses of these circuits not only benefit fundamental research, but also, they underlie applications in biotechnology and synthetic biology for the precise control of gene expression. Two-component systems (TCS) are the predominant signal-transduction pathways by which bacteria and several other organisms process environmental stimuli and convert them into intracellular gene-expression responses (Buschiazzo and Trajtenberg, 2019; Parkinson and Kofoid, 1992). In their canonical form, TCSs comprise as one component a membrane-spanning, homodimeric sensor histidine kinase (SHK) that relays signals to the cell interior. SHKs generally consist of N-terminal extracellular/periplasmic sensor and C-terminal cytosolic effector modules, the latter of which subdivide into the DHp (dimerization and phospho-accepting histidine) and CA (catalytic) domains. In signal-dependent manner, SHKs undergo autophosphorylation at a conserved, eponymous histidine residue to enable subsequent phosphoryl-group transfer to an aspartic acid residue within the response regulator (RR), i.e., the second TCS component. Once phosphorylated, the RR commonly dimerizes and is thereby rendered capable of sequence-specific DNA binding and transcription initiation at target promoters (Gao et al., 2019). As it is the overall levels of phosphorylated RR that govern the physiological output of the TCS, many SHKs are bifunctional and also exert phosphatase in addition to their kinase activities (Möglich et al., 2009; Russo and Silhavy, 1993). Exceedingly stringent and steep regulatory responses can be achieved by finely balancing the opposing RR phosphorylation and dephosphorylation reactions. Depending on the signal, one or the other of the two elementary activities prevails, and the SHK consequently acts as either net kinase or net phosphatase. SHKs are specific for their respective RRs, thus allowing the coexistence in a single cell of multiple TCSs with minimal crosstalk (Laub and Goulian, 2007).

Among the plethora of known TCSs, certain representatives respond to light as their cognate stimulus (Möglich, 2019; Ohlendorf and Möglich, 2022). Pertinent TCSs comprise soluble photoreceptor SHKs that reside in the cytosol and absorb light at specific wavelengths within the near-ultraviolet (UV) to near-infrared (NIR) range of the electromagnetic spectrum (Krell et al., 2010; Möglich, 2019). Beyond their pivotal physiological role, these TCSs are of prime interest in biotechnology as they unlock the control of gene expression by light, an approach belonging to optogenetics (Deisseroth et al., 2006). As an inducer of gene expression, light brings several benefits in that it affords superior spatial and temporal precision; can be applied, dosed, and withdrawn with ease and in automatable fashion; and penetrates living tissue to a wavelength-dependent extent (Weissleder, 2001). Several light-responsive TCSs have become available and support strong regulatory responses covering the electromagnetic spectrum from the near-UV to the NIR (Lazar and Tabor, 2021; Ohlendorf and Möglich, 2022). Apart from little-modified naturally occurring setups (Tabor et al., 2011), additional photoreceptor SHKs have been engineered by modular recombination of sensor and effector entities (Levskaya et al., 2005; Möglich et al., 2010, 2009). As a case in point, the widely deployed pDusk and pDawn plasmids resort to the blue-light-inhibited SHK YF1 originating from the fusion of a light-oxygen-voltage (LOV) photosensor to the DHp/CA effector of the *Bradyrhizobium japonicum* FixL SHK (Möglich et al., 2009; Ohlendorf et al., 2012). Irrespective of the wealth of setups for the optical control of bacterial gene expression, there is a dearth of circuits affording stringent responses to NIR light (Ong et al., 2018).

We previously expanded the pDusk plasmid family by exchanging said LOV photosensor for photosensory core modules (PCM) of bacterial phytochromes (BphP) (Meier et al., 2024a; Multamäki et al., 2022). The resultant pREDusk systems respond to red rather than blue light which facilitates multiplexing (Meier et al., 2024b; Multamäki et al., 2022) with scores of bacterial optogenetic circuits that mostly sense blue light (Ohlendorf and Möglich, 2022). As an added benefit, red light penetrates mammalian tissue to a higher extent than blue light owing to reduced scattering and absorption by cellular constituents, hemoglobin, and other pigments (Ash et al., 2017; Weissleder, 2001). BphPs react to red and NIR light via biliverdin (BV) chromophores covalently attached within their PCMs (Davis et al., 1999; Hughes et al., 1997; Takala et al., 2020). The linear tetrapyrrole BV traverses between the *Z* and *E* isomers of its C15=C16 double bond which underpin the red-absorbing Pr and the far-red-absorbing Pfr states of BphPs, respectively. Red and NIR light triggers the photochromic *Z*/*E* isomerization of BV and thereby drives the photoconversion in the Pr → Pfr and Pfr → Pr directions, respectively. In conventional BphPs, as in those underpinning the pREDusk systems, the Pr state is the thermodynamically more stable form prevailing in darkness. After red-light exposure, these BphPs revert to their Pr resting state thermally within the dark-recovery reaction which is generally protracted and plays out over many hours to days. Conventional BphPs therefore effectively act as ratiometric red/NIR-light sensors. In common with many natural BphP-SHKs (Huber et al., 2024; Karniol and Vierstra, 2003; Lamparter et al., 2002; Multamäki et al., 2021), the engineered ones in the pREDusk circuits exhibit net kinase activity towards their RR within their Pr states and net phosphatase activity within their Pfr states. In pREDusk, red light therefore downregulates target gene expression by several hundred-fold (Meier et al., 2024a; Multamäki et al., 2022). More recently, we inverted the light response of the underlying BphP-SHKs by changing the composition of the linkers through which their sensor and catalytic modules transduce signals (Meier et al., 2024a). Doing so yielded BphP-SHKs with net kinase activity in their Pfr states and net phosphatase activity in their Pr states, thus supporting pronounced increases of gene expression under red light in the resultant pDERusk systems.

A second group of bacterial phytochromes, denoted bathy-BphPs, adopt the Pfr rather than the Pr state in darkness (Karniol and Vierstra, 2003). Consequently, NIR light promotes the Pfr → Pr conversion, as does visible light, if to a lesser degree. To the extent it has been studied, the thermal dark recovery in bathy-BphPs is generally fast and complete within seconds to minutes. Due to these aspects, bathy-BphPs are considered to primarily serve as sensors of light vs. darkness rather than of a specific light color (Huber et al., 2024).

Informed by spectroscopic analyses, we here harness the PCMs of select bathy-BphPs for stringently controlling TCS output and bacterial gene expression by NIR light. The resultant plasmid systems, dubbed pNIRusk, prompt stringent upregulation of gene expression under NIR light. Equally advantageously and unexpectedly, the engineered pNIRusk circuits show only minimal activation by red light. To enable multiplexed applications, we also advance light-regulated setups based on orthogonal TCSs. The pNIRusk plasmids not only afford pronounced NIR-light responses in *Escherichia coli* laboratory strains but also in the probiotic Nissle bacteria and in *Agrobacterium tumefaciens*. Taken together, we deliver versatile implements for the precision control of gene expression by NIR light in both fundamental and applied research.

Results

Bathy-Bacteriophytochromes as Sensors of Light and pH

To subject bacterial gene expression to NIR light, we considered the PCMs of previously reported bathy-BphPs as input modules (Rottwinkel et al., 2010; Tasler et al., 2005). To better inform our choice, we expressed, purified, and spectroscopically characterized BphP PCMs from *Azorhizobium caulinodans* (*Ac*PCM), *Agrobacterium vitis* (*Av*PCM), and *Pseudomonas aeruginosa* (*Pa*PCM). Consistent with earlier reports, all three PCMs fully populated the Pfr state in darkness with distinct Soret- and Q-band absorbance peaks at 415 nm and (755 ± 3) nm, respectively (**Fig. 1a-c**). Exposure to NIR light drove the complete conversion to the Pr state characterized by maxima of the Soret band at (393 ± 2) nm and of the Q band at around (698 ± 3) nm. Illumination with blue and red light prompted partial population of the Pr state to around 65-80% and 40-70%, respectively. Strikingly, at pH 8.0, the Q-band intensity of the pure Pr state differed markedly across the PCMs, with *Pa*PCM showing the strongest signal, and *Ac*PCM the weakest. Based on earlier reports, these data may principally reflect differences in BV protonation at the pyrrole C ring (Borucki et al., 2005; Zienicke et al., 2013). To put this notion to the test, we recorded absorbance spectra for the three PCMs at pH values between 6 and 11 (**Fig. 1d-f**). While the Q-band intensity remained invariant within the Pfr state, in the Pr state it increased with decreasing pH. By evaluating the corresponding absorbance peak heights as a function of pH, we determined apparent pK_a values of 7.2 ± 0.1, 8.0 ± 0.1, and 9.9 ± 0.2 for *Ac*PCM, *Av*PCM, and *Pa*PCM, respectively (**Fig. 1g**). Following previous studies (Zienicke et al., 2013), we tentatively ascribe these pK_a values to (de-)protonation of the nitrogen atom within the pyrrole C ring in BV. In addition, pH also affects the photostationary Pr:Pfr ratio under red light which increases with rising alkalinity (**Fig. 1h**, **Suppl. Fig. 1a-c**).

Having ascertained the bathy-character of the candidate PCMs and their Pr⇄Pfr interconversion, we next assessed the kinetics of reversion to their dark-adapted Pfr state at pH 8.0 following NIR illumination (**Fig. 1i**). All three PCMs fully recovered to the Pfr state in single- exponential fashion, but the time constants τ for *Ac*PCM and *Av*PCM of (140 ± 1) s and (80 ± 1) s were markedly lower than the value of (610 ± 1) s for *Pa*PCM. Closely similar recovery time constants resulted after exposing the samples to red instead of NIR light (**Suppl. Fig. 1d-e**). With rising alkalinity, we observed slower and less complete recovery after NIR-light exposure (**Suppl. Fig. 1g-l**).

Potentially, this deceleration directly owes to BV deprotonation evidenced in the absorbance spectra. That is because BV remains protonated in the Pfr state across all investigated pH values, but within the Pr state it is progressively deprotonated at higher pH values. The return to the Pfr state hence requires re-protonation at elevated pH values which may slow down the dark recovery.

Subjecting Bacterial Gene Expression to NIR-Light Control

To achieve control of bacterial expression by NIR light, we adapted the previous *Dm*REDusk plasmid which encodes a red-light-responsive chimeric SHK originating from the fusion of the PCM of *Deinococcus maricopensis* BphP (*Dm*BphP) to the DHp/CA fragment of *B. japonicum* FixL (Meier et al., 2024a). As not least seen in the above spectroscopic experiments, bathy-BphPs also respond to visible besides NIR light, which stands to cause undesired crosstalk with genetic circuits that react to, e.g., blue or red light. We reasoned that this adverse effect may be ameliorated by deliberately rendering the NIR-responsive systems comparatively insensitive to light. Guided by the above spectroscopic analyses, we selected *Ac*PCM and *Av*PCM as they exhibit a faster Pr→Pfr dark recovery than *Pa*PCM, which translates into a lower effective light sensitivity at photostationary state (Ohlendorf and Möglich, 2022; Ziegler and Möglich, 2015).

With *Ds*Red as a fluorescence reporter gene, we replaced *Dm*PCM within *Dm*REDusk by either *Ac*PCM or *Av*PCM, while maintaining the same register for the linker between the PCM and DHp/CA moieties (**Suppl. Fig. 2**). In the resultant plasmids, referred to as pNIRusk, the heme oxygenase enzyme for intracellular BV provision, the BphP-SHK, and the matching RR FixJ thus form a constitutively expressed tricistronic operon (**Fig. 2a**). The initial pNIRusk designs, denoted ’0’, showed constant reporter fluorescence, largely independent of illumination, despite their BphP- SHKs adhering to the same design as in *Dm*REDusk (**Fig. 2b**).

Spurred by our recent construction of the pDERusk setups (Meier et al., 2024a), we hypothesized that light responsiveness may be installed by varying the composition of the linker between the PCM and DHp/CA moieties. Using the PATCHY approach (Ohlendorf et al., 2016), we generated libraries of the *Ac*NIRusk and *Av*NIRusk setups that differ in the length and sequence of said linker. By assessing the reporter fluorescence in darkness and under NIR and red light, we

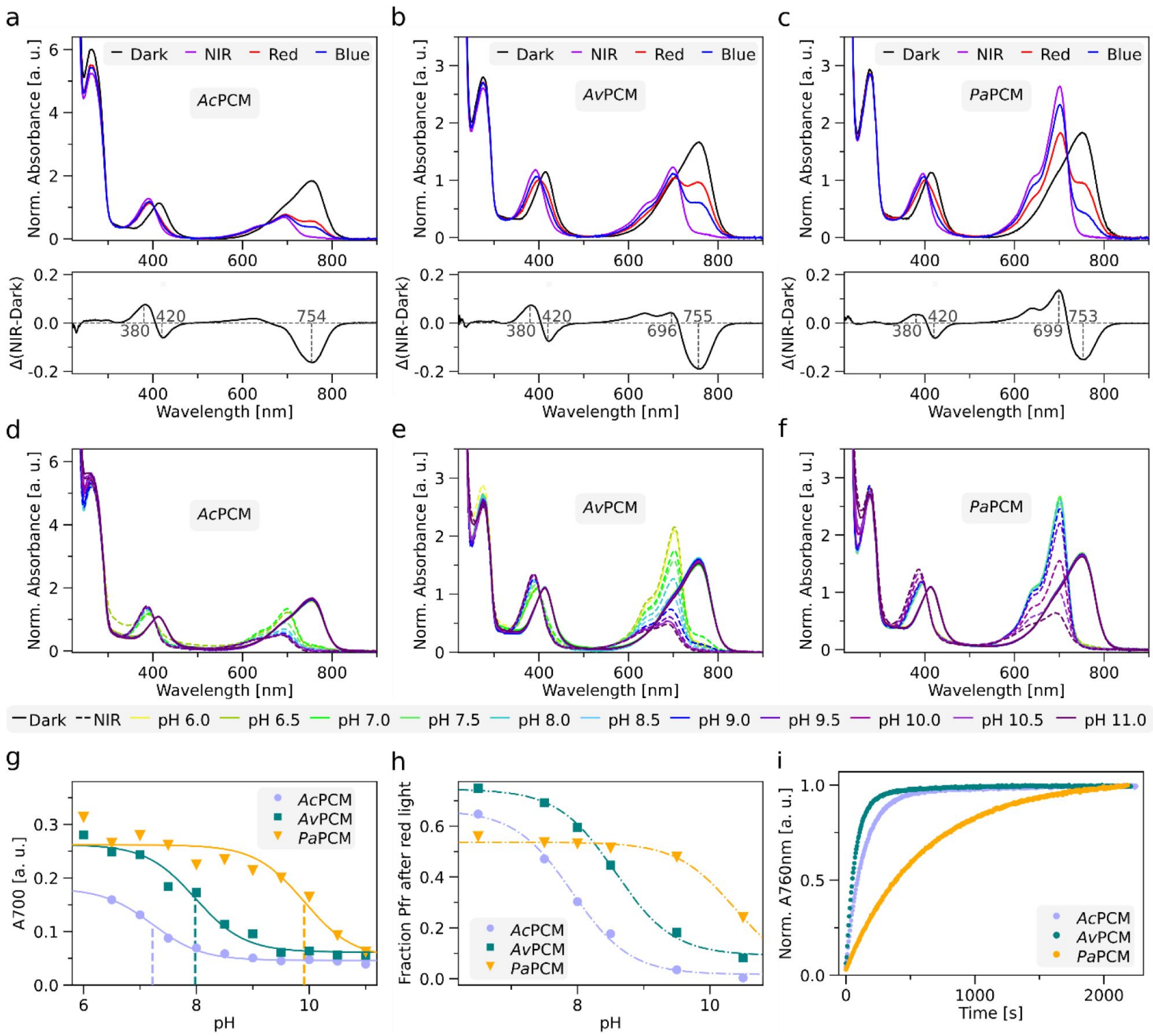


Figure 1.

Spectroscopic characterization of bacteriophytochrome photosensory core modules (PCM) from *Azorhizobium caulinodans* (*Ac*PCM), *Agrobacterium vitis* (*Av*PCM), and *Pseudomonas aeruginosa* (*Pa*PCM).

(a) UV/vis absorbance spectra of *Ac*PCM at pH 8 in darkness (black line), and after illumination with near-infrared (NIR, purple line), red (red), and blue light (blue). Data are normalized to the absorbance at 403 nm within the Soret band. The lower panel shows the NIR-dark difference absorbance spectrum with maxima and minima marked. (b) and (c) as in (a) but for *Av*PCM and *Pa*PCM. (d) Absorbance spectra of *Ac*PCM in darkness (solid lines) and after NIR illumination (dashed lines) at different pH values as indicated by color. (e) and (f) as in (d) but for *Av*PCM and *Pa*PCM. (g) Absorbance at 700 nm for *Ac*PCM (light purple circles), *Av*PCM (dark green squares), and *Pa*PCM (orange triangles) upon NIR-light exposure at different pH values. Data were fitted to the Henderson-Hasselbalch equation to determine apparent pK_a values (dashed lines, *Ac*PCM: 7.2 ± 0.1, *Av*PCM: 8.0 ± 0.1, *Pa*PCM: 9.9 ± 0.2). (h) Calculated Pfr fractions of *Ac*PCM (light purple circles), *Av*PCM (dark green square), and *Pa*PCM (orange triangles) after red-light exposure at different pH values. The underlying absorbance spectra are shown in **Suppl. Fig. 1a-c**. (i) Dark-recovery kinetics at 25°C and pH 8 after NIR-illumination as followed by absorbance at 760 nm. Time constants for *Ac*PCM (light purple), *Av*PCM (dark green), and *Pa*PCM (orange) were determined by single-exponential fits.

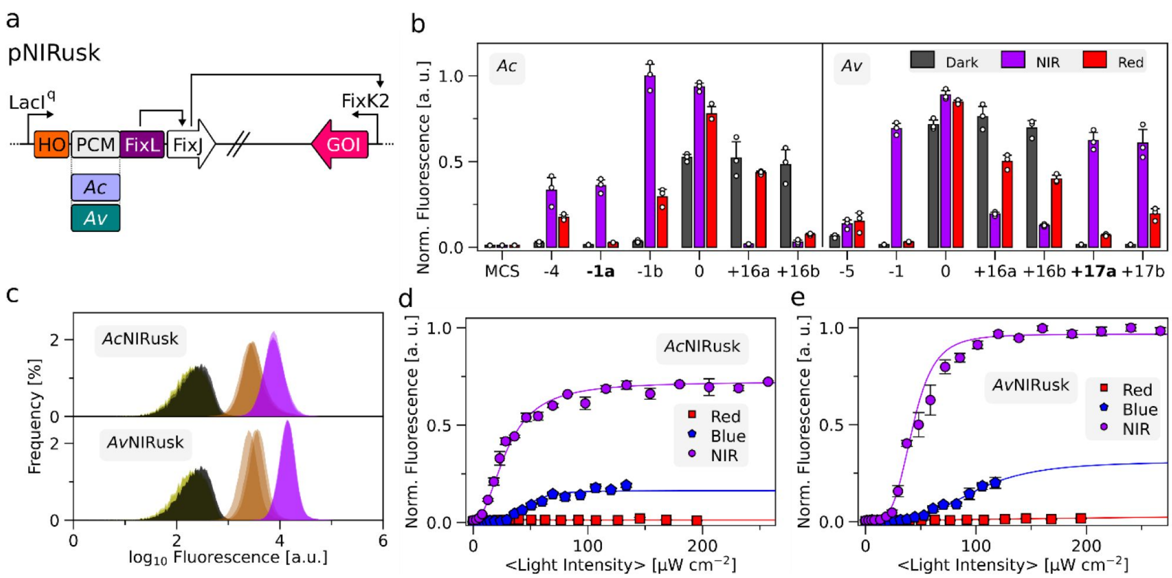


Figure 2.

Engineering of light-regulated sensor histidine kinases (SHK) based on photosensory core modules (PCM) from bathy-bacteriophytochromes (BphP).

(a) Schematic of the pNIRusk plasmids to control gene expression in bacteria by NIR light. The LacI^q promoter constitutively expresses a tricistronic operon consisting of heme oxygenase (HO1), the light-regulated SHK (chimera of *Ac*/*Av*PCM and FixL), and the response regulator FixJ. When phosphorylated, FixJ binds to the FixK2 promoter and triggers the expression of a gene of interest (GOI). (b) Response of select *Ac*NIRusk (left) and *Av*NIRusk (right) systems to different light conditions (grey: darkness; purple: NIR light; red: red light). *Ds*Red reporter fluorescence was normalized to the optical density of the bacterial cultures. Individual pNIRusk variants are denoted by the relative lengths of the linkers within their underlying SHKs (see **Suppl. Fig. 3**), with the variants chosen for further analyses highlighted in bold. As control, an empty-vector construct (MCS) is shown, and data are normalized to the maximal fluorescence. (c) Flow-cytometric analyses of bacteria containing *Ac*NIRusk-1a (top) or *Av*NIRusk+17a (bottom) upon incubation in darkness (black), moderate NIR (brown), or strong NIR light (purple) (also see **Suppl. Fig. 5**). An empty-vector control (MCS) is shown in green. (d) Light-dose response of bacteria containing *Ac*NIRusk-1a incubated in NIR (purple circles), red (red squares), and blue light (blue pentagons). (e) as in (d) but for *Av*NIRusk+17a.

identified five light-switchable variants for *Ac*NIRusk and six for *Av*NIRusk (**Fig. 2b**, **Suppl. Fig. 3**). We refer to individual variants by their linker lengths relative to the initial constructs. Although the light- responsive variants all rely on the same PCMs and DHp/CA modules, they drastically differed in their response to NIR light, with some elevating and others lowering reporter-gene expression compared to darkness. We subsequently focused on variants inducing gene expression under NIR light, albeit with intriguing variation in their response to red light. For instance, *Ac*NIRusk-1b with a linker one residue shorter than in *Dm*REDusk showed sizeable induction under red light. By contrast, the variant *Ac*NIRusk-1a with the same linker length but different sequence did much less so. For an eventual application, we deem a reduction of red-light-induced activation advantageous. We thus continued with *Ac*NIRusk-1a and *Av*NIRusk+17a, both of which strongly induce expression under NIR light compared to darkness but only show small responses to red light. For simplicity, we refer to these setups as *Ac*NIRusk and *Av*NIRusk in the following.

Flow cytometry revealed homogeneous single-cell fluorescence distributions for *E. coli* bacteria harboring *Ac*NIRusk or *Av*NIRusk (**Fig. 2c**). When incubated in darkness, the median fluorescence was only 1.2-fold above the background fluorescence. Saturating NIR light (800 nm, 250 μW cm^{-2}) induced largely uniform shifts of the entire cell population to several 10-fold higher median fluorescence for *Ac*NIRusk and *Av*NIRusk, respectively, an effect similar in magnitude to the red-light response observed previously for the pDERusk systems (Meier et al., 2024a). At sub-saturating light doses, the single-cell fluorescence of bacteria containing *Ac*NIRusk and *Av*NIRusk uniformly rose to intermediate levels with increasing NIR-light intensity (**Suppl. Fig. 4**-**5**).

At the ensemble level, the average fluorescence per cell density ramped up sigmoidally with NIR light by up to around 70- and 150-fold for *Ac*NIRusk and *Av*NIRusk, respectively (**Fig. 2d-e**). The two systems exhibited half-maximal activation at relatively high NIR-light intensities of (29 ± 1) μW cm^{-2} and (43 ± 1) μW cm^{-2}, i.e., about an order of magnitude above the half-maximal red-light doses previously determined for induction of the pREDusk and pDERusk systems (Meier et al., 2024a; Multamäki et al., 2022). Arguably, these differences reflect at least in part the respective dark- recovery times of the underlying bathy-BphPs which are in the seconds to minutes range rather than in the hours range for conventional BphPs.

Red light also induced *Ac*NIRusk and *Av*NIRusk but only around 2- and 6-fold relative to darkness (**Fig. 2d-e**). Blue light elicited a stronger 16- and 25-fold induction, respectively, at the highest probed light intensity of around 120 μW cm^{-2}. Higher blue-light intensities were not accessed owing to instrumental limitations and onsetting phototoxicity. From the data, we determined half-maximal blue-light doses of (46 ± 3) μW cm^{-2} and (100 ± 20) μW cm^{-2} for *Ac*NIRusk and *Av*NIRusk, respectively. Given the around 1.7-fold higher energy of blue compared to NIR light, the circuits therefore trigger at similar half-maximal photon fluencies for either light color. At least qualitatively, the degree of induction by the individual light colors (red, blue, NIR) follows the same trend as the photochemical responses evidenced in the underlying *Ac*PCM and *Av*PCM (see **Fig. 1a-c**) where NIR light elicited the most pronounced Pr formation, followed by blue and red light.

Engineering Orthogonal Light-responsive Two-Component Systems

In common with earlier members of the pDusk pedigree, the pNIRusk systems are based on the FixLJ TCS from *B. japonicum* (Anthamatten and Hennecke, 1991). Combinations of several of these setups would hence experience crosstalk at the level of RR phosphorylation which, in all but a few cases, is undesired. To overcome this limitation, we set out to engineer light-responsive TCSs orthogonal to those used in pDusk, pNIRusk, and related setups. As recently surveyed, several TCSs have been characterized to the point that they have found application in synthetic biology (Lazar and Tabor, 2021). Based on their reported stringency of regulation, we selected as candidate TCSs TtrSR from *Shewanella baltica* that responds to tetrathionate and TodST from *Pseudomonas putida* that can detect toluene compounds (Daeffler et al., 2017; Lau et al., 1997).

Starting from *Dm*REDusk, we substituted the original FixL DHp/CA domains for those from TtrS, while also exchanging FixJ for the TtrR RR and putting the expression of a *Ds*Red reporter under control of the Pttr promoter that is subject to regulation by TtrSR (**Fig. 3a**). Within this framework, we then varied the linker interconnecting the *Dm*PCM and the TtrS DHp/CA module via PATCHY (Ohlendorf et al., 2016). Screening of the resultant linker library for differential *Ds*Red expression under darkness, and NIR and red light identified several variants, termed *Dm*Ttr, with substantial light responses (**Fig. 3a**, **Suppl. Fig. 6a**). Notably, the overall expression levels within the identified *Dm*Ttr systems were markedly higher than those in the pREDusk and pNIRusk setups.

Next, we extended the strategy to *Av*NIRusk by introducing TodST. Of note, TodS is a complex SHK with integrated phosphorelay that comprises in sequential order PAS-1, DHp-1, CA-1, REC, PAS- 2, DHp-2, and CA-2 domains (Busch et al., 2009). Toluene signals, detected by PAS-1, modulate the activity of the DHp-1/CA-1 unit which phosphorylates the REC domain that in turn governs the activity of the DHp-2/CA-2 module and eventual phosphorylation of the TodT RR. Earlier reports showed that a miniaturized TodS variant only comprising the PAS-1, DHp-2, and CA-2 domains, dubbed mini-TodS, retains toluene sensitivity and stringent regulation of TodT (Silva-Jiménez et al., 2012). Consequently, we replaced the FixLJ TCS in *Av*NIRusk by the TodS DHp-2/CA-2 fragment, while also introducing the TodT RR and its cognate PtodX promoter (**Fig. 3b**). Linker variation by PATCHY and screening for *Ds*Red reporter fluorescence pinpointed several variants, termed *Av*Tod, which either elevated or lowered gene expression in response to NIR light (**Fig. 3b**, **Suppl. Fig. 6b**).

For further analyses, we selected the variants *Dm*Ttr+7b, *Av*Tod+16, and *Av*Tod+21 with PCM-DHp/CA linkers 7, 16, and 21 residues, respectively, longer than that in the parental *Dm*REDusk circuit. Exposed to red light, *Dm*Ttr+7b elicited an about 15-fold decrease of gene expression compared to darkness (**Suppl. Fig. 7a**). Whereas *Av*Tod+16 supported a 20-fold reporter-fluorescence decrease under NIR light, *Av*Tod+21 promoted an 8-fold increase of gene expression (**Suppl. Fig. 7b-c**). Although the attained regulatory efficiencies pale in comparison to those of *Dm*REDusk (Meier et al., 2024a) and *Av*NIRusk, the data show that the principal design and screening strategies readily translate to other TCSs.

To allow multiplexing of several light-responsive circuits in a single bacterial cell, we modified the *Dm*DERusk and *Av*NIRusk plasmids by replacing their kanamycin resistance marker, their ColE1 origin of replication (ori), and their *Ds*Red reporter gene by a streptomycin marker, the CloDF13 ori, and the yellow-fluorescent YPet reporter, respectively. The resultant *Dm*DERusk-Str-YPet and *Av*NIRusk-Str-YPet circuits showed similar responses to red and NIR light as the parental plasmids they derive from (**Suppl. Fig. 7d-e**). Pairs of compatible plasmids with different TCSs, oris, resistance markers, and fluorescent reporters were transformed into bacteria, followed by cultivation in darkness, under red (100 µW cm^{-2}) or NIR light (200 µW cm^{-2}), or under both light colors. By employing individual plasmid combinations, the expression of the *Ds*Red and YPet reporters could be individually controlled by the two light colors (**Fig. 3c**, **Suppl. Fig. 7f**). For instance, when combining *Dm*DERusk-Str-YPet with *Av*Tod+21-*Ds*Red, YPet and *Ds*Red expression rose under red and NIR light, respectively, whereas the joint application of both light colors induced both reporter genes. Taken together, well-chosen plasmid sets enable the orthogonal expression control of two target genes by red and NIR light. Notably, the plasmids can be deployed as is with no further modifications required. Rather than using saturating light intensities as in the above experiments, yet finer-grained expression control can be exerted by adjusting the red- and NIR-light intensities.

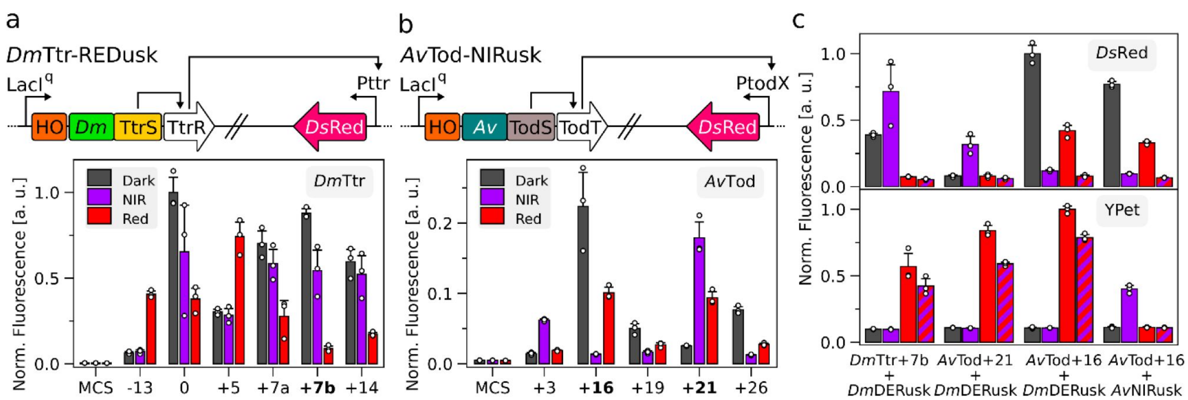


Figure 3.

Engineering of orthogonal light-regulated two-component systems (TCS). (a) Schematic of the *Dm*Ttr-REDusk circuit based on the TtrSR TCS from *Shewanella baltica* combined with the photosensory core module (PCM) from the *Deinococcus maricopensis* bacteriophytochrome (BphP) (top).

Reporter fluorescence of bacteria bearing circuits with different linker lengths of their underlying sensor histidine kinases (SHK) upon cultivation in darkness (grey), NIR (purple), and red light (red) (bottom). An empty-vector control is denoted as 'MCS', and data are normalized to the maximal fluorescence. Variants are marked by the relative lengths of the linkers within their underlying SHKs compared to pREDusk (see **Suppl. Fig. 6**). Variants characterized further are highlighted in bold. (b) as in (a) but for the *Av*Tod-NIRusk circuit based on the TodST TCS from *Pseudomonas putida* combined with the BphP PCM from *Agrobacterium vitis*. (c) Multiplexing of bacteria containing two optogenetic circuits, as noted below the graph, to separately control the expression of *Ds*Red (top) and YPet (bottom) reporters. After incubation in darkness (gray), NIR (purple), red (red), and combined NIR and red light (red and purple stripes), fluorescence was measured and normalized to the maximal value.

Applications in the Laboratory and Beyond

As TCSs widely recur across bacterial phyla, they often retain functionality upon transfer between different bacterial strains. Earlier reports not least demonstrated that representatives of the pDusk family support light-regulated gene expression across several prokaryotes (Ohlendorf and Möglich, 2022). For instance, pDusk and its derivative pDawn (Ohlendorf et al., 2012) have been successfully deployed in *E. coli* Nissle 1917 (*Ec*N) and other probiotic bacteria (Cui et al., 2021; Magaraci et al., 2016; Pan et al., 2021; Sankaran and del Campo, 2019; Yang et al., 2020). These experiments augur the application of light-responsive bacteria in the gut of mammals as programmable, living therapeutics (Ohlendorf and Möglich, 2022). To date, however, pertinent applications are limited by the shallow tissue penetration of blue light required for triggering pDusk and pDawn. Although upconverting nanoparticles can be administered jointly with the light-responsive bacteria to allow actuation by NIR light, this approach introduces complexity and departs from a fully self-contained, optogenetic strategy (Yang et al., 2020). To assess whether the presently generated light-responsive gene-expression systems may overcome this grave limitation, we introduced *Ac*NIRusk and *Av*NIRusk into *Ec*N. Both setups mediated a more than 100-fold upregulation of reporter-gene expression under NIR light with half-maximal doses of (27 ± 1) and (28 ± 2) μW cm^{-2}, respectively (**Fig. 4a**).

We next probed the performance of *Av*NIRusk and *Dm*REDusk in the biotechnologically relevant *Agrobacterium tumefaciens*. To this end, we constructed derivative plasmids, denoted *Av*NIRlux and *Dm*REDlux, that subject the expression of a luciferase operon under the control of the light-responsive TCSs (**Fig. 4b**). After conjugational transfection into *A. tumefaciens*, the light response was assessed by spotting the bacteria on agar plates (**Fig. 4c**). As visualized by luminescence, the *Av*NIRlux system strongly induced expression upon incubation under NIR light compared to darkness. Different from *E. coli* (see **Fig. 2e**, **Suppl. Fig. 8a**), red light activated to similar extents as NIR light, whereas blue light induced expression only weakly. In the case of *Dm*REDlux, the expression of the *lux* reporter was strongly reduced under both blue and red light, but not under NIR light or in darkness. Although *A. tumefaciens* harbors two endogenous BphPs (Karniol and Vierstra, 2003), our experiments did not detect any differences in growth or phenotype between cultivation under NIR and red light. We next characterized the response of *A. tumefaciens* cultured in liquid medium. Once reaching the stationary phase after around 20 h (**Suppl. Fig. 8b**), the cultures bearing *Av*NIRlux exhibited a 7-fold activation of *lux* expression by NIR light compared to darkness. For *Dm*REDlux, the *lux* expression was around 300-fold higher in darkness than under red light (**Fig. 4d**, **Suppl. Fig. 8c-d**).

Discussion

Bathy-Bacteriophytochromes as Tunable Sensors of Light Intensity and Quality

As conventional bacteriophytochromes do, bathy-BphPs transition between their Pr and Pfr states with 15*Z*- and 15*E*-configured biliverdin, respectively. In marked contrast to the conventional representatives, bathy-BphPs assume their 15*E*-configured Pfr state as the thermodynamically most stable form in darkness, rather than their Pr state. Despite this principal difference, the overall PCM architecture and the residues lining the bilin binding pocket are remarkably conserved across known BphPs. Notwithstanding limited successes in reprogramming certain BphPs (Böhm et al., 2025; Xu et al., 2024), the molecular determinants underlying bathy-character remain elusive and their design demanding.

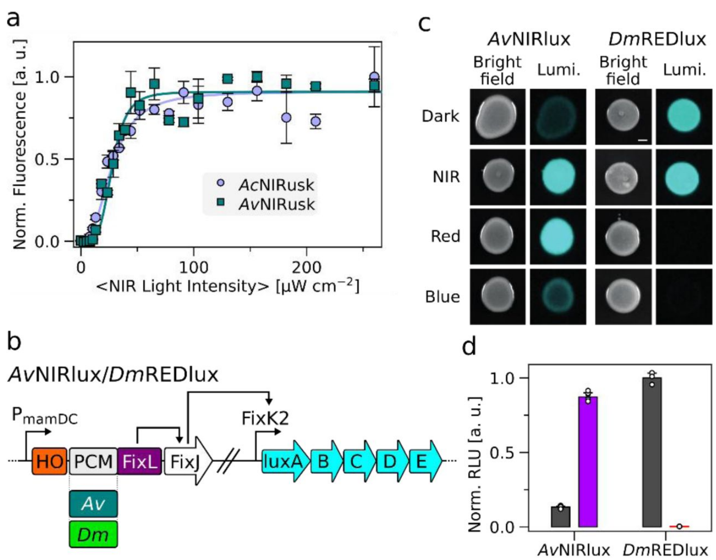


Figure 4.

Light-regulated gene expression in *Escherichia coli* Nissle (*Ec*N) and *Agrobacterium tumefaciens*.

(a) Light-dose response of *Ec*N harboring *Ac*NIRusk (light purple circles) and *Av*NIRusk (dark green squares) after incubation at varying NIR-light intensities. (b) Schematic of the *Av*NIRlux and *Dm*REDlux circuits based on *Av*PCM and *Dm*PCM. The tricistronic cassette comprising heme oxygenase (HO1), the sensor histidine kinase (chimera of *Av*/*Dm*PCM and FixL), and FixJ is controlled by the constitutive PmamDC promoter from *Magnetospirillum gryphiswaldense*, whereas the FixK2 promoter regulates the expression of the *luxABCDE* operon. (c) *A. tumefaciens* bacteria bearing *Av*NIRlux (left) and *Dm*REDlux (right) were spotted on plates and incubated at different light conditions (darkness, NIR, red, or blue light), followed by luminescence (Lumi.) measurements (see **Suppl. Fig. 8a**). (d) *A. tumefaciens* bacteria carrying *Av*NIRlux and *Dm*REDlux were incubated in liquid culture for 20 h under different light conditions (gray: darkness, purple: NIR light, red: red light), followed by recording of reporter luminescence (see **Suppl. Fig. 8c-d**).

Owing to their Pfr dark-adapted state, bathy-BphPs undergo net Pfr→Pr conversion when irradiated with light within the near-UV to NIR region of the electromagnetic spectrum. Put another way, there is no light quality that would fully populate the dark-adapted Pfr state. Bathy-BphPs surmount this challenge by a fast thermal Pr→Pfr recovery that restores the Pfr state within seconds to minutes. The primary physiological role of bathy-BphPs may hence be that of sensors of light quantity rather than quality (i.e., color) (Huber et al., 2024).

Intriguingly, we uncovered a profound pH influence on photoreception in bathy-BphPs. With rising alkalinity, the absorption cross-section of the Q band strongly decreases within the Pr state but remains unchanged for Pfr. Based on earlier studies (Borucki et al., 2005; Velazquez Escobar et al., 2015; Zienicke et al., 2013), we attribute these spectral changes to (de-)protonation of the nitrogen atom of the BV pyrrole C within the Pr state. The pertinent pK_a values range between 7-8 for *Ac*PCM and *Av*PCM, similar to the value of 7.6 for the bathy-BphP Agp2 from *A. tumefaciens* (Zienicke et al., 2013), but substantially lower than the value of about 10 for *Pa*PCM. Inspection of the experimentally determined *Pa*PCM and Agp2PCM structures in their Pfr states (Schmidt et al., 2018; Yang et al., 2008) and the corresponding Pfr models for *Ac*PCM and *Av*PCM (Abramson et al., 2024) reveals highly conserved chromophore-binding pockets with most residues adjacent to BV identical in all four proteins (**Suppl. Fig. 9**). However, *Pa*PCM features tyrosine and serine residues at positions 190 and 275 rather than phenylalanine and alanine, respectively, in the equivalent places of the other three PCMs. Although speculative at present, these systematic differences may render the *Pa*PCM binding pocket comparatively hydrophilic and thereby account for the divergent protonation equilibrium of the pyrrole C ring.

BV deprotonation within the Pr state at elevated pH incurs at least three principal effects that under constant light all promote population of the Pr state at the cost of the Pfr state. First, the Pr→Pfr dark recovery slows down between pH 7 and 10 for all three investigated PCMs. At least in part, this may reflect the need for BV re-protonation upon return from the Pr to the Pfr state which remains fully protonated at all investigated pH values. Second, as noted above, BV deprotonation leads to a drastic reduction of the Q-band absorbance within the Pr state (see **Fig. 1**). For a given intensity of visible light, photoconversion to the Pfr state thus becomes less effective, thereby again favoring the Pr state at photostationary state. Third, BV deprotonation also goes along with a dramatic decrease in the apparent photoconversion quantum yields in the Pr→Pfr vs. the Pfr→Pr directions (**Suppl. Fig. 1f**). As best exemplified for *Ac*PCM, at pH 6.5, the respective quantum yields for the photoreversible Pr⇄Pfr interconversion are about equal, but at pH 10.5, the quantum yield in the Pr→Pfr direction is less than 1% that in the Pfr→Pr direction. Together with the pH-dependent reduction in the Pr-state Q-band absorbance, this mechanism leads to vastly different red-light- induced photostationary states as a function of pH (see **Fig. 1h**). Whereas at low pH, saturating red light results in about 60-70% Pfr population, at high pH values almost no Pfr state remains (**Fig. 5**). As a direct consequence, at pH 10.5, red light counter-intuitively drives the complete conversion to the Pr state of *Ac*PCM. Put another way, in its deprotonated form this bathy-BphP becomes essentially colorblind and acts as a pure light-intensity sensor. Although less pronounced, *Av*PCM and *Pa*PCM follow the same trend with pH.

While it is debatable whether the rhizobia *Ac*PCM and *Av*PCM derive from ever encounter such extreme pH swings, it has been proposed that the resultant changes in photochemical properties could serve a sensing function (Zienicke et al., 2013). Even independent of this aspect, our present analyses show how pH impacts bathy-BphP photoreception at multiple levels. While it remains true that all light colors net promote the Pfr→Pr transition, they do so to different extents which can have a profound impact on downstream signal transduction as discussed below. We propose that modulation of the BV protonation midpoint has been harnessed during evolution to sculpt the light response of extant bathy-BphPs according to the individual needs of the respective organism of origin. In line with this hypothesis, we presently evidence a considerable span of pK_a values for BV protonation across several bathy-BphP PCMs.

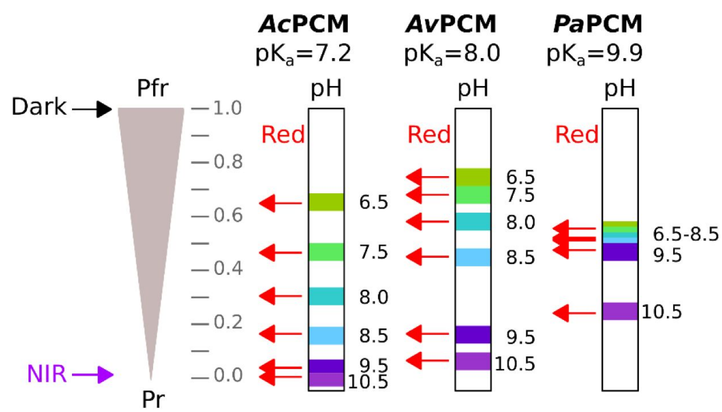


Figure 5.

Photoreception by select bathy-bacteriophytochromes (BphP) at different pH values.

Throughout the entire tested pH range, the photosensory core modules (PCM) of the bathy-BphPs from *Azorhizobium caulinodans* (*Ac*PCM), *Agrobacterium vitis* (*Av*PCM), and *Pseudomonas aeruginosa* (*Pa*PCM) adopt a pure Pfr state in darkness with fully protonated biliverdin (BV) chromophore. NIR light drives the complete conversion to the Pr state in which BV can be either protonated or deprotonated as governed by pH and the respective pK_a value of the bathy-BphP. Red light populates a photostationary mixture with Pr and Pfr proportions strongly depending on pH. At high pH values, red light can drive the bathy-BphP nearly completely to the Pr state for *Ac*PCM and *Av*PCM. Owing to a higher pK_a value for BV protonation, *Pa*PCM is less affected but follows the same trend.

Engineering of NIR-Light-Responsive Proteins

As best exemplified by sensor histidine kinases, PCMs of bathy-BphPs can occur in conjunction with the same type of effector as those of conventional BphPs (Blum et al., 2025). However, it has been unclear whether a single, specific output protein can be regulated in its activity by both PCM types. The successful recombination of bathy-BphP PCMs with the same DHp/CA module earlier controlled by conventional BphP PCMs (Meier et al., 2024a; Multamäki et al., 2022) now demonstrates the principal validity of this idea. At least to some extent, the PCMs of conventional and bathy-BphPs are hence interchangeable, a finding that indicates shared or compatible signal-transduction mechanisms, as also suggested by the conserved structure and photochemistry across the BphP class (Möglich, 2019; Takala et al., 2020). As one important ramification, numerous previously engineered red-light-responsive receptors may thus be subjugated to NIR-light control by swapping out their underlying PCMs from conventional BphPs for ones from bathy-BphPs (Chernov et al., 2017; Etzl et al., 2018; Leopold et al., 2022, 2019; Levskaya et al., 2005; Schmidl et al., 2014; Stabel et al., 2019; Stüven et al., 2019; Tang et al., 2021). Compared to the challenging approach of imparting bathy-character via limited PCM modifications (Böhm et al., 2025; Xu et al., 2024), the design task thus considerably simplifies.

We identify several caveats that likely hold for future design endeavors, too. Notably, the connection between PCM and effector of choice needs to be conducive to transmitting light signals. When at present employing the same linker register that supports red-light-inhibited gene expression in *Dm*REDusk, we initially derived *Ac*PCM- and *Av*PCM-based systems giving rise to constitutive target gene expression without pronounced light responses. Adjusting the length and sequence of the linker intervening the PCM and the DHp/CA domains then yielded variants with robust gene-expression responses to NIR light. Based on structural and biochemical evidence, said linker is expected to form a continuous parallel α-helical coiled coil within the homodimeric SHK (Bódizs et al., 2024; Burgie et al., 2024; Diensthuber et al., 2013; Jacob-Dubuisson et al., 2018; Malla et al., 2024; Möglich, 2019; Möglich et al., 2009). Although other BphP-based photoreceptor designs may involve linkers adopting different structures, the variation and probing of length and sequence variants generally represents a worthwhile strategy that may eventually bear fruit. As, depending on the use case, sizeable variant libraries may have to be assessed, two principal aspects emerge that benefit the implementation of this strategy. First, the construction of libraries covering the desired diversity in linker composition should be efficient, which can, e.g., be achieved by resorting to the PATCHY protocol (Ohlendorf et al., 2016) as done here. Second, the screening for light-regulated function should afford sufficient throughput, e.g., by coupling photoreceptor function to a fluorescent readout or cell survival.

Tunability of NIR-Light-Activated Two-Component Systems

The recombination of bathy-BphP PCMs with a SHK effector module led to the pNIRusk plasmids for stringent activation by NIR light of bacterial gene expression. As the underlying PCMs absorb light over the entire near-UV to NIR region of the electromagnetic spectrum, the pNIRusk systems are expected to also activate when exposed to other light colors. To dampen the commonly undesirable activation by visible light and thereby facilitate the multiplexing with other photoresponsive circuits, we deliberately chose *Ac*PCM and *Av*PCM because of their comparatively fast Pr→Pfr thermal recovery. At photostationary state under constant illumination, a faster recovery results in an effective desensitization against light, be it of blue, red, or NIR color (Ziegler and Möglich, 2015). Consistent with this design rationale, the *Ac*NIRusk and *Av*NIRusk setups indeed exhibit half-maximal light doses around ten times higher than for, e.g., *Dm*DERusk which relies on a conventional BphP PCM (Meier et al., 2024a). Given its low phototoxicity, NIR light may be applied at even high powers without harming the biological system under study. We note that NIR light around 800 nm is commonly used in a therapeutic setting for photobiomodulation at intensities exceeding the ones presently used by up to several orders of magnitude (Hamblin, 2016).

Perplexingly, the *Ac*NIRusk and *Av*NIRusk circuits respond moderately or very weakly to blue and red light, respectively (see **Fig. 2d, e**). Although the strength of the gene-expression response monotonically increases with the extent of Pfr→Pr photoconversion elicited by the individual light colors within the underlying PCMs (see **Fig. 1a-c**), there is no direct proportionality. For instance, *Ac*PCM roughly assumes a 0.7:0.3 photostationary Pr:Pfr equilibrium under red light (see **Fig. 1a**), but at the level of gene expression, the response of the *Ac*NIRusk circuit to red light is only 2% that of the NIR-light response. Initially unexpected, this baffling effect therefore further benefits the application of the pNIRusk circuits in combination with other light-responsive entities, be it optogenetic circuits or fluorescent reporters (Meier et al., 2024b; Tabor et al., 2011).

Our work provides pointers to the potential molecular underpinnings in that we isolated additional pNIRusk variants with similar NIR-light responses but substantially higher expression under red light (see **Fig. 2b**). Notably, these variants only differ in the composition of their linkers connecting PCM to effector. Informed by our recent investigation of linker variants that culminated in the generation of *Dm*DERusk (Meier et al., 2024a), we hypothesize that the different red-light responses find their origin in an altered balance between the counteracting elementary histidine kinase and phosphatase activities of SHKs. At the ensemble level, even small perturbations, e.g., caused by linker variation or changes in the Pr:Pfr proportion, can utterly tilt the net activity in either the kinase or phosphatase direction (Landry et al., 2018; Möglich, 2019; Möglich et al., 2009; Russo and Silhavy, 1993). The histidine kinase-phosphatase duality of TCSs thus provides a means of rendering the signal response to light highly malleable and adaptable. At a practical level, when engineering light-responsive TCSs, we advise to screen multiple candidate variants that may well differ substantially in their signal response. As noted previously (Meier et al., 2024a), similar concepts may also operate for cyclases (GGDEF) and phosphodiesterases (EAL) enzymes that frequently occur in conjunction with BphP PCMs (Gourinchas et al., 2017) and that antagonistically catalyze the making and breaking, respectively, of the bacterial second messenger cyclic-di-guanosine monophosphate (Jenal et al., 2017).

Applications in the NIR Future

The advent of stringently controlled NIR-responsive bacterial gene-expression platforms now facilitates a number of innovative applications. First, the currently available optogenetic setups for expression regulation in bacteria predominantly respond to blue and red light but are often blind to NIR light (Ohlendorf and Möglich, 2022). Given their fully intended relative insensitivity and their unexpectedly weak response to visible light, the pNIRusk systems lend themselves to combinations with pertinent setups. For example, applied at low to moderate intensity, blue light would trigger one setup, e.g., pAurora2 or EL222-based systems (Jayaraman et al., 2016; Ranzani et al., 2024), but leave pNIRusk largely untouched. By contrast, NIR light would selectively activate pNIRusk as the LOV receptors underlying the blue-light-responsive setups are insensitive to this light quality. To facilitate multiplexed applications of pNIRusk, we supply derivatives reliant on alternative TCSs that can be readily combined in single bacterial cells and mediate the expression of several target genes orchestrated by two light colors.

Second, NIR light penetrates mammalian tissue much more deeply than light of shorter wavelengths (Lehtinen et al., 2022; Weissleder, 2001). This opens up intriguing applications involving programmable, light-responsive bacteria as agents inside the bodies of mammals, e.g., as living therapeutics (Cubillos-Ruiz et al., 2021; Hartsough et al., 2020). The pDawn system has already been applied in this capacity within probiotic bacteria, but the approach has been hampered by delivering blue light *in situ* (Cui et al., 2021; Magaraci et al., 2016; Pan et al., 2021; Sankaran and del Campo, 2019; Yang et al., 2020). One remedy has been the use of upconverting nanoparticles (UCNP) administered alongside the light-responsive bacteria (Pan et al., 2021; Yang et al., 2020). Notably, the UCNPs can be charged by absorption of several photons in the NIR range to then emit a photon in the blue spectral region. Though feasible, pertinent applications introduce complexity and depart from a purely genetically encoded optogenetic strategy. The new pNIRusk setups directly address this challenge as they can be activated by essentially the same wavelengths previously used for charging the UCNPs. Towards this goal, we demonstrate at present that both *Ac*NIRusk and *Av*NIRusk operate in the probiotic *E. coli* Nissle strain out of the box, without need for optimization. Used in place of pDawn, the pNIRusk systems thus obviate the need for the UCNPs and thereby broaden the application scope of living therapeutics considerably.

Third, we demonstrate functionality of pNIRusk and pREDusk in the biotechnologically relevant *A. tumefaciens* (**Fig. 4c-d**) which is widely used as a transfection vehicle in plant biotechnology (Liu et al., 2013). Prospectively, both pNIRusk and pREDusk may achieve spatiotemporal control of *A. tumefaciens* infectivity, e.g., to confine plant transfection to specific time windows or organs. Of particular advantage, the bathochromically shifted absorbance spectra of BphPs compared to plant Phys (Hughes and Winkler, 2024; Yi et al., 2025) stand to enable NIR-light control without interfering with plant photoreception and photoautotrophic growth.

Methods

Molecular Biology

Oligonucleotides and plasmids used in this study are listed in **Suppl. Tables 1** and **2**. All plasmid constructs were confirmed by Sanger sequencing (Microsynth Seqlab, Göttingen, Germany). To produce bathy-BphP PCMs, the respective genes were introduced into a pCDF-Duet plasmid (Novagen) already containing heme oxygenase 1 (HO1) from *Synechocystis* sp. for biliverdin supply (Mukougawa et al., 2006; Stüven et al., 2019). To this end, the genes encoding the PCMs from *A. caulinodans* (Uniprot A8HU76, residues 1-501), *A. vitis* (Uniprot B9JR96, residues 1-499), and *P. aeruginosa* (Uniprot Q9HWR3, residues 1-497) were amplified by PCR from the template plasmids pQX091, pQX086, and pQX037, respectively (Xu et al., 2024), followed by Gibson assembly (Gibson et al., 2009).

The initial *Ac*NIRusk and *Av*NIRusk constructs (denoted '0') were created by exchanging *Dm*PCM in *Dm*REDusk (Meier et al., 2024a) for *Ac*PCM and *Av*PCM, respectively (**Suppl. Fig. 2**), via PCR amplification and Gibson assembly. The second amino acid in *Ac*PCM and *Av*PCM was changed from proline to threonine to maintain the polycistronic structure of the operon comprising HO1, the BphP-SHK, and the RR FixJ. To apply the PATCHY strategy (Ohlendorf et al., 2016), start constructs were generated by elongating the PCM-DHp/CA linkers in the initial *Ac*NIRusk and *Av*NIRusk plasmids as reported before (Meier et al., 2024a). The resulting constructs thus comprise the N-terminal portions of the parental BphPs from A. *caulinodans* (Uniprot A8HU76) and *A. vitis* (Uniprot B9JR96) up to and including residues 520 and 518, respectively, connected to the C-terminal fragment of *B. japonicum* FixL starting from residue 257 (Uniprot P23222). At the junction, a restriction site for *Ksp*AI and a one-nucleotide frameshift were inserted (**Suppl. Fig. 3**). Cloning was performed by two consecutive overhang-PCRs and blunt-end ligation.

The TtrSR TCS from *Shewanella baltica* was introduced into *Dm*REDusk (Meier et al., 2024a) by first replacing the FixK2 promoter by the 85-bp minimal promoter sequence PttrB185-269 (Daeffler et al., 2017) via PCR amplification and restriction cloning using *Bgl*II and *Xba*I sites. Next, the *B. japonicum* FixLJ TCS was exchanged for the *S. baltica* TtrSR (Genbank CP000891.1, loci Sbal195_3858 and Sbal195_3859) via Gibson assembly using as a PCR template a synthetic gene (Gene Art) with codons optimized for expression in *E. coli*. For conducting PATCHY, a start construct was made that includes residues 1-539 of the *Dm*BphP (Uniprot E8U3T3) and the *S. baltica* TtrSR TCS (Uniprot A9L3S5) starting at residue 407. Cloning was performed as described above except that an *Eco*32I restriction site and a two-nucleotide frameshift were placed at the junction between the two gene fragments (**Suppl. Fig. 6a**).

The cognate TodX promoter of the TodST TCS from *Pseudomonas putida* was amplified by PCR from pMIR77 (Ramos-González et al., 2002) and inserted into pDusk (Ohlendorf et al., 2012) by restriction cloning with *Bgl*II and *Xba*I. Next, the TodST TCS was amplified from pMIR66 (Ramos-González et al., 2002) and fused to the *Bacillus subtilis* YtvA LOV domain within pDusk via Gibson cloning. The *Av*Tod-NIRusk PATCHY start construct was obtained by replacing YtvA-LOV with HO1 and and *Av*PCM (residues 1-518) via Gibson assembly. The resultant construct comprised the TodS starting from residue 726 (Uniprot E0X9C7) and TodT. At the junction, a *KspA*I restriction site and a one-nucleotide frameshift were inserted (**Suppl. Fig. 6b**).

For multiplexing experiments, the *Ds*Red Express2 reporter gene (Strack et al., 2008) in *Dm*DERusk and *Av*NIRusk was replaced by the YPet reporter gene (Nguyen and Daugherty, 2005). Additionally, the kanamycin (Kan) resistance marker and the ColE1 origin of replication (ori) were exchanged for streptomycin (Str) and a CloDF13 ori via Gibson assembly. A version of *Av*NIRusk containing a multiple-cloning site rather than a fluorescent reporter was cloned by Gibson assembly and deposited at Addgene for distribution (accession number 235084).

To apply *Dm*REDusk and *Av*NIRusk in *Agrobacterium tumefaciens*, the *Photorhabdus luminescence luxABCDE* operon from pBAM2-luxAE (Dziuba et al., 2021) was cloned into the miniTn5 delivery plasmid pBAM2 (Uebe, unpublished). The sequence was amplified via PCR, digested with *Nde*I/*Sac*I, and ligated into a similarly digested and Fast-AP-dephosphorylated pBAM2 vector to generate pBAM2-luxAE. Subsequently, the light-regulated TCS cassettes of *Dm*REDusk (*DrhemO-Dmpcm-fixL*) and *Av*NIRusk (*DrhemO-Avpcm-fixL*) were amplified. These genes were set under the control of the strong constitutive *M. gryphiswaldense* PmamDC promoter. The *B. japonicum* FixK2 promoter was also amplified by PCR. The amplified fragments were fused by overlap extension PCR and digested with *Nde*I/*Not*I as was the pBAM2-luxAE. Ligation resulted in the reporter plasmids *Dm*REDlux and *Av*NIRlux in which the *luxABCDE* operon is under the control of the FixK2 promoter.

Protein Purification

For production of *Ac*PCM, *Av*PCM, and *Pa*PCM, *E. coli* LOBSTR cells (Andersen et al., 2013) were transformed with the respective pCDF-Duet-PCM/HO1 constructs. Gene expression and protein purification were performed as before (Meier et al., 2024a). Briefly, two flasks with 800 mL lysogeny broth (LB) medium each supplemented with 100 µg mL^{-1} Str were inoculated with a 5-mL overnight culture, and bacteria were grown in baffled Erlenmeyer flasks at 37°C and 225 rpm agitation. At an optical density at 600 nm (OD_{600}) of around 0.7, the expression was induced by 1 mM isopropyl-β-thiogalactopyranoside (IPTG), and 0.5 mM δ-aminolevulinic acid was added to support biliverdin incorporation. The expression was performed at 16°C and 225 rpm agitation in darkness for around 65 h. All following steps were performed in green safe light. After harvesting the cells by centrifugation, they were resuspended in lysis buffer [50 mM tris(hydroxy-methyl)-amino-methane (Tris)/HCl, 20 mM NaCl, 10 mM imidazole, 1 mM β-mercaptoethanol, pH 8.0, protease inhibitor completeOne (Roche), 1 mg mL^{-1} lysozyme]. Cells were lysed by sonification, and the supernatant of the centrifuged lysate was applied to a HiTrap TALON crude column loaded with cobalt-nitrilotriacetic resin (Cytiva). The column was washed with buffer [50 mM Tris/HCl, 20 mM NaCl, 1 mM β-mercaptoethanol, pH 8.0], and the His-tagged proteins were eluted by an imidazole gradient from 0 to 0.5 M. Fractions were analyzed by denaturing SDS polyacrylamide gel electrophoresis (PAGE) in the presence of 1 mM Zn^{2+} to visualize biliverdin incorporation (Berkelman and Lagarias, 1986) and pooled based on protein content. During an overnight dialysis at 4°C into buffer [50 mM Tris/HCl, 20 mM NaCl, 1 mM β-mercaptoethanol, pH 8.0], the His-tag was cleaved off by His_6-tagged TEV protease. The sample was applied twice to a gravity-flow IMAC column with HisPur cobalt resin (ThermoFisher) using the same buffers as before. Cleavage of the His-tag and protein purity were analyzed by Zn-SDS-PAGE. Fractions were pooled, dialyzed overnight at 4°C against storage buffer [20 mM Tris/HCl, 20 mM NaCl, 10% (w/v) glycerol, 1 mM dithiothreitol, pH 8.0], concentrated via spin filtration, and stored at -80°C.

UV/vis Absorption Spectroscopy

Absorption spectroscopy was carried out on a Cary60 UV/vis spectrophotometer (Agilent Technologies) at 25°C under green safe light. Protein samples were diluted into buffers with pH ranging from 6.0 to 11.0 (30 mM 2-(N-morpholino)ethanesulfonic acid/HCl, 30 mM NaCl, pH 6.0/6.5/7.0; 30 mM Tris/HCl, 30 mM NaCl, pH 7.5/8.0/8.5/9.0; 30 mM glycine/HCl, 30 mM NaCl, pH 10.0/10.5/11.0] (Rumfeldt et al., 2019). Samples were illuminated for 120 s with LEDs emitting in the NIR [(800 ± 18) nm], red [(650 ± 10) nm] and blue [(470 ± 5) nm]. Data were normalized to the Soret absorbance at 403 nm. Pr→Pfr dark recovery kinetics were monitored every 5 s at 760 nm and fitted to single-exponential functions using the Fit-o-mat software (Möglich, 2018). Pr:Pfr ratios after red or blue illumination were calculated via linear combination of the basis spectra as described previously (Yi et al., 2025). The variation of photochemical parameters with pH was evaluated by fitting to the Henderson-Hasselbalch equation. The ratio of the quantum yields Φ_p and Φ_q for the red-light-driven photoconversion in the Pr→Pfr and Pfr→Pr directions, respectively (see **Suppl. Fig. 1f**), can be determined as follows (Yi et al., 2025):

$$\Phi_q/\Phi_p = P_R/P_{FR} \times (1 - 10^{-A_{Pr}})/(1 - 10^{-A_{Pfr}})$$

P_{R} and P_{FR} are the Pr and Pfr proportions at red-light-induced photostationary state, and A_{Pr} and A_{Pfr} denote the absorbance in the Pr and Pfr states at the peak emission wavelength of the red-light source.

Generation and Screening of Linker Variants

*Ac*NIRusk, *Av*NIRusk, *Dm*Ttr-REDusk and *Av*Tod-NIRusk linker variants were created with the PATCHY strategy (Meier et al., 2024a; Ohlendorf et al., 2016). Sets of forward and reverse primers to generate linker variants were designed using a custom Python script (*https://github.com/vrylr/PATCHY*). In brief, PCR amplification was carried out using the primer sets in a final concentration of 0.25 µM. After purifying the PCR product by gel extraction, the start constructs were digested with *Ksp*AI or *Eco*32I (**Suppl. Fig. 3**, 6), followed by phosphorylation with T4 polynucleotide kinase and ligation with T4 DNA ligase. The reaction mixes were transformed into DH10b cells and plated on LB plates supplemented with kanamycin (LB/Kan). Plates were incubated overnight at 37°C in darkness, or under red light [100 µW cm^{-2}, (650 ± 11) nm] or NIR light [200 µW cm^{-2}, (800 ± 18) nm]. Applied light intensities were calibrated with a power meter (model 842-PE, equipped with a 918D-UV-OD3 silicon photodiode, Newport, Darmstadt, Germany). Bacterial colonies developing red color upon overnight incubation, indicative of *Ds*Red production, were used to inoculate 200 µL LB/Kan in a clear 96-well microtiter plate (MTP) (Nunc, ThermoFisher). Bacteria were incubated for 21-24 h at 30°C and 750 rpm agitation. Next, the cultures were diluted 100-fold into fresh LB/Kan medium and split into three separate cultures which were then incubated at 37°C and 750 rpm agitation in either darkness, red light [100 µW cm^{-2}, (650 ± 11) nm], or NIR light [200 µW cm^{-2}, (800 ± 18) nm]. After 18 h, the optical density at (600 ± 9) nm (OD_{600}) and the *Ds*Red fluorescence [(554 ± 9) nm excitation and (591 ± 20) nm emission] were measured with a Tecan Infinite M200 Pro MTP reader (Tecan Group, Ltd., Männedorf, Switzerland). The *Ds*Red fluorescence was normalized to OD_{600} and represent the mean ± s.d. of three biologically independent replicates.

Light-Dose-Response Assays in *Escherichia coli*

Analyses were performed as before (Meier et al., 2024a; Multamäki et al., 2022). *E. coli* CmpX13 or *E. coli* Nissle (*Ec*N, DSMZ no. 115365) bacteria containing the respective construct were used to inoculate 5 mL LB medium supplemented with 50 µg mL^{-1} Kan or 100 µg mL^{-1} Str (LB/Str). Cultures were incubated for 21-24 h at 30°C and 225 rpm agitation under non-inducing light conditions. Under green safe light, the bacteria were diluted 100-fold into fresh LB/Kan or LB/Str, before distributing 200 µL each into 64 wells of a black-walled MTP with clear bottom (µClear, Greiner BioOne, Frickenhausen, Germany). As a control, bacteria harboring an MCS variant were also placed on the MTP. After sealing the MTP with a gas-permeable membrane, it was placed on an Arduino-controlled illumination device. For the application of red and blue light, a previous setup was used (Hennemann et al., 2018) with LED peak wavelengths of (624 ± 8) nm and (463 ± 12) nm, respectively. To apply NIR light, we devised a new matrix of 8-by-8 LEDs emitting at (800 ± 18) nm (LED800-01AU, Roithner Lasertechnik) which is controlled by an Arduino circuit board (**Suppl. Fig. 4**). The bacteria inside the MTPs were incubated for 18 h at 37°C and 750 rpm agitation. The illumination was applied in 1:2 or 1:10 duty cycles (i.e., 1 s light and 1 s darkness, or 20 s light and 180 s darkness). After incubation, the OD_{600} and *Ds*Red fluorescence were determined as above. For measurements of YPet fluorescence, excitation and emission wavelengths of (500 ± 9) nm and (554 ± 9) nm were used. Following normalization to OD_{600}, the MCS background was subtracted. Data represent mean ± s.d. of three biological replicates and were evaluated as a function of the light intensity averaged over the duty cycle according to Hill isotherms using Fit-o-mat.

Flow Cytometry

E. coli CmpX13 cells (Mathes et al., 2009) containing *Ac*NIRusk, *Av*NIRusk, or an empty-vector control (MCS) were analyzed by flow cytometry (Meier et al., 2024a). After overnight incubation for 21-24 h at 30°C and 750 rpm agitation, bacterial cultures were diluted 100-fold into fresh LB/Kan medium. For each variant, the cultures were split in three and incubated for 18 h at 37°C and 750 rpm agitation in either darkness, red light [100 µW cm^{-2}, (650 ± 11) nm], or different intensities (18-270 µW cm^{-2}) of NIR light [(800 ± 18) nm]. Following incubation, the cultures were diluted 10-fold into phosphate-buffered saline (1x sheath fluid, Bio-Rad) and analyzed on a S3e cell sorter (Bio-Rad). The *Ds*Red fluorescence of at least 200,000 events was detected using excitation lasers at 488 nm and 561 nm and an emission channel at (585 ± 15) nm. Data were fitted to skewed Gaussian distributions using Fit-o-mat. Experiments were repeated three times with similar results.

Multiplexing of Optogenetic Circuits

Pairs of plasmids harboring light-responsive TCSs were serially transformed into chemically competent *E. coli* CmpX13. 200 µL LB supplemented with 50 µg mL^{-1} Kan and 100 µg mL^{-1} Str (LB/Kan+Str) were inoculated with single bacterial clones and incubated for 21-24 h inside MTP wells at 30°C and 750 rpm agitation in darkness. The cultures were then diluted 100-fold into LB/Kan+Str before dispensation of 200 µL each into individual wells of four µClear MTPs. The MTPs were incubated for 18 h at 37°C and 750 rpm agitation in either darkness, under red light [100 µW cm^{-2}, (624 ± 8) nm], NIR light [200 µW cm^{-2}, (800 ± 18) nm], or both light colors combined. Where applicable, red light was applied continuously from below using the Arduino-controlled RGB LED matrix, whereas NIR light shone from above. After 18 h incubation, the OD_{600}, and *Ds*Red and YPet fluorescence were measured as above. Data were normalized to the maximal *Ds*Red and YPet readings and represent mean ± s.d. of three biological replicates.

Luciferase Assays in Agrobacterium tumefaciens

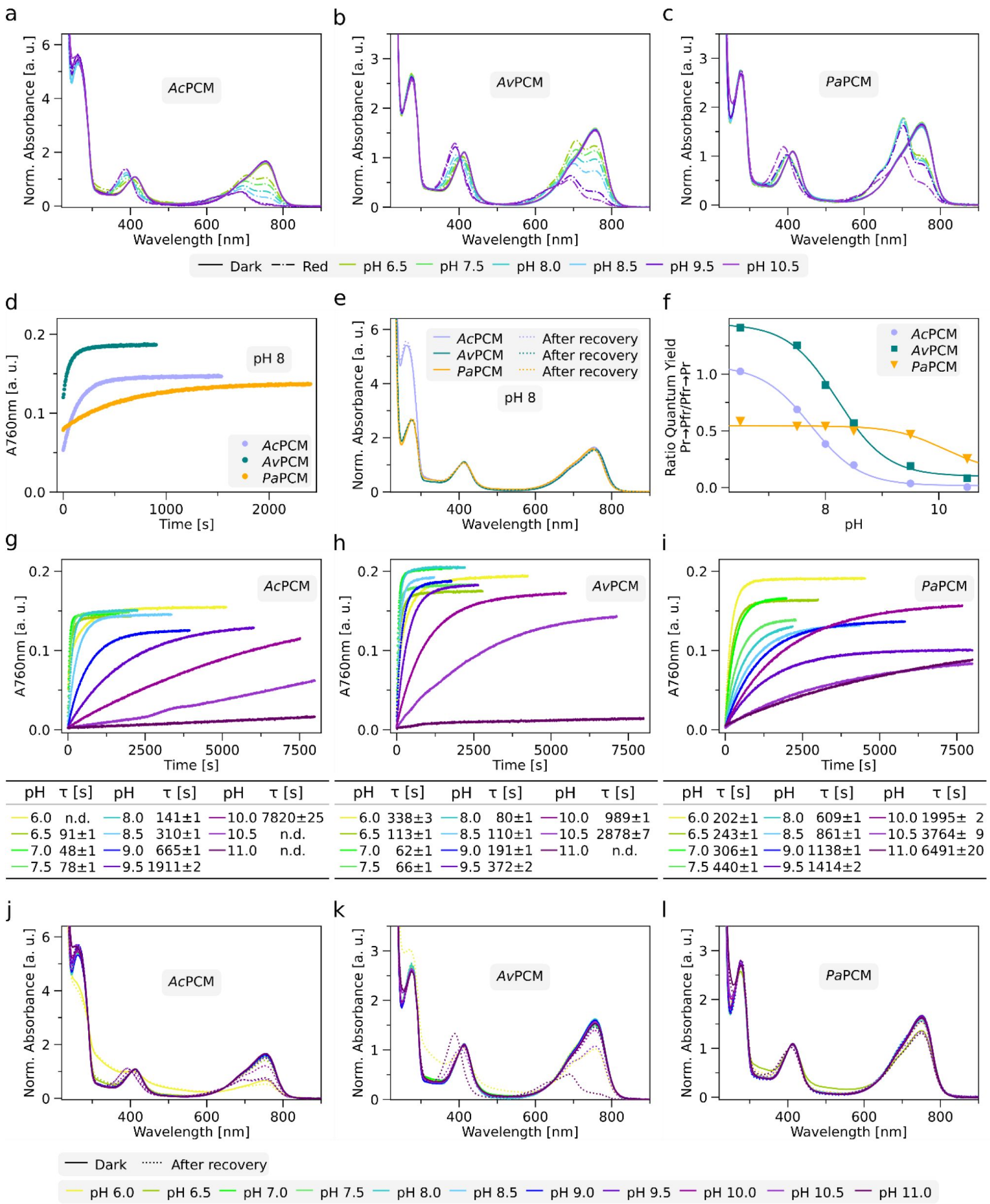
The *Dm*REDlux and *Av*NIRlux plasmids were transferred to *Agrobacterium tumefaciens* C58 (also known as *Agrobacterium fabrum* C58, Lab collection, ATCC 33970) by biparental conjugation using *E. coli* WM3064 as a diaminopimelic acid auxotrophic donor strain (William Metcalf, UIUC, unpublished; *thrB1004 pro thi rpsL hsdS lacZΔ*M15 RP4-1360 *Δ(araBAD)567 ΔdapA1341::[erm pir (wt)])*. Unless specified otherwise, C58 was routinely cultivated in LB medium at 28°C with 170 rpm agitation. Selection for mini-Tn5 insertion mutants carrying the *lux* reporter constructs was carried out on solid LB with 1.5% (w/v) agar and 50 µg mL^{-1} Kan. *E. coli* WM3064 strains carrying plasmids were cultivated at 37°C with 200 rpm shaking in LB supplemented with 0.1 mM DL-a,e-diaminopimelic acid (DAP) and 25 µg mL^{-1} Kan or 50 µg mL^{-1} ampicillin (Amp).

The protocol for biparental conjugation was described before (Dziuba et al., 2021). Briefly, cultures of the C58 receptor and plasmid-containing *E. coli* strains were cultivated in LB or LB/Amp medium overnight. Then, 2×10^9 cells of each strain were mixed in 15-mL falcon tubes and pelleted at 5,421 g for 10 min. The supernatant was discarded while the cell pellet was resuspended in 100-200 µL LB medium and spotted onto LB agar plates. After overnight incubation at 28°C, cells were resuspended in 1 mL LB and transferred to 15-mL falcon tubes containing 9 mL LB medium. After incubation for 2 h at 28°C and 120 rpm agitation, the cell suspension was pelleted by centrifugation, and the supernatant was discarded. Following resuspension in 1 mL LB medium, 100 µL of 10- and 100-fold diluted cell suspensions were plated on LB-Kan agar plates. After incubation for three days at 28°C, Kan-resistance plasmid insertion mutants were transferred to fresh LB/Kan agar plates and screened by PCR.

At least three randomly selected transconjugants containing the luciferase reporter constructs were analyzed for luminescence. Each experiment was performed in duplicate. Luminescence signals, measured as arbitrary light units (ALU), were detected using a Tecan Infinite M200 Pro MTP reader equipped with a luminometer module (Tecan Group, Ltd., Männedorf, Switzerland) during culture growth in LB medium at 23°C and 280 rpm. Measurements were taken every 20 minutes over 300 cycles (~90 h). ALU readings were normalized to the optical density at 650 nm (OD_{650}) to calculate the relative light units (RLU). To this end, tested clones were inoculated into 600 µL LB/Kan medium in 24-well plates and incubated for ~ 20 h under non-inducing conditions (i.e., red light for *Dm*REDlux and darkness for *Av*NIRlux, respectively) at 28°C and 140 rpm. Cultures were then diluted 10-fold into 1 mL LB/Kan and incubated under identical conditions for an additional 4-5 h. Subsequently, the OD_{650} was adjusted to 0.1 by adding fresh LB/Kan medium, and 100 µL of the normalized cell suspension was transferred into two 96-well MTPs. Cultivation of both MTPs was initiated simultaneously, with one incubated under dark conditions and the other one under red [(650 ± 11) nm] or NIR light [(800 ± 18) nm].

Alternatively, luminescence was assessed using a spotting assay. Precultures were prepared as above, and 5 µL of the OD_{650}-normalized cell suspension were spotted onto LB/Kan agar plates. The plates were incubated for 44 h at 28°C under dark conditions or under blue [(470 ± 13) nm], red [(650 ± 11) nm], or NIR light [(800 ± 18) nm]. Luminescence signals were documented as cumulative signal intensities using the Bio-Rad ChemiDoc XRS+ System (Bio Rad Laboratories Inc., Herkules, USA), with images captured every 0.6 s over a 1-min period. In addition, colorimetric images were acquired to visualize the outlines of the spotted cell suspensions. Luminescence was quantified from the image time series by calculating the slope of the mean pixel intensities within the outlined cell suspension areas using the Image J Fiji package (v1.54f) (Schindelin et al., 2012). Background correction was performed by subtracting the mean pixel intensity of an uninoculated region of the agar plate.

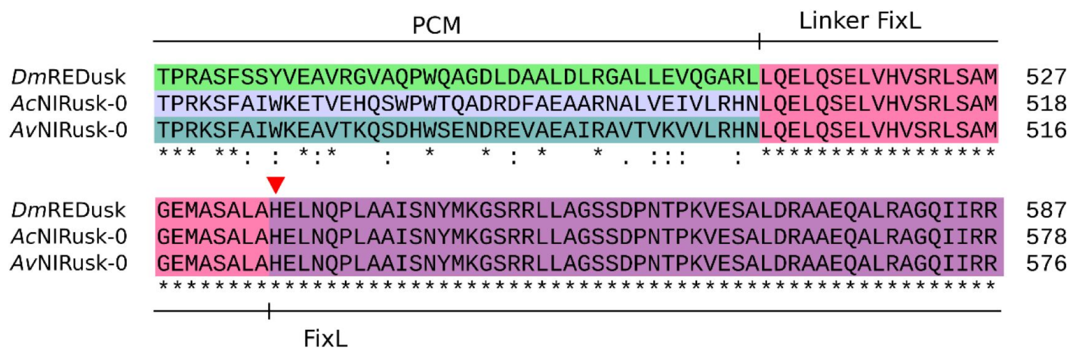
Supplementary Information



Supplementary Figure 1.

Additional spectroscopic characterization of bacteriophytochrome photosensory core modules (PCM) from *Azorhizobium caulinodans* (*Ac*PCM), *Agrobacterium vitis* (*Av*PCM), and *Pseudomonas aeruginosa* (*Pa*PCM).

(a) Absorbance spectra of *Ac*PCM at different pH values as indicated below the graph in darkness (solid line) and after red-light-illumination (dotted line). Data are normalized to the absorbance at 403 nm within the Soret band. (b) and (c) as in (a) but for *Av*PCM and *Pa*PCM. (d) Dark-recovery kinetics of *Ac*PCM (light purple circles), *Av*PCM (dark green circles), and *Pa*PCM (orange circles) monitored at 760 nm, 25°C, and pH 8 after illumination with red light. Time constants determined by single-exponential fits are (147 ± 1) s, (71 ± 1) s and (599 ± 2) s for *Ac*PCM, *Av*PCM, and *Pa*PCM, respectively. (e) Absorbance spectra recorded in darkness (solid lines) and after dark recovery (dotted lines) for *Ac*PCM (light purple), *Av*PCM (dark green), and *Pa*PCM (orange) at pH 8. (f) Ratio of the quantum yields for Pr→Pfr over Pfr→Pr photoconversion as a function of pH. (g) Dark-recovery kinetics of *Ac*PCM recorded at 760 nm after illuminating with NIR light at 25°C and different pH as indicated below the graph. Time constants were determined by single-exponential fits, with the results shown in the table. (h) and (i) as in (g) but for *Av*PCM and *Pa*PCM. (j) Absorbance spectra of *Ac*PCM recorded in darkness (solid lines) and after dark recovery (dotted lines) at different pH as indicated below the graph. (k) and (l) as in (j) but for *Av*PCM and *Pa*PCM.

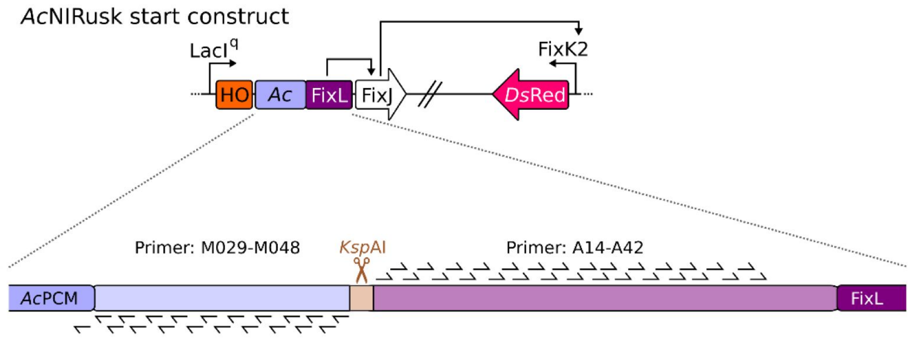


Supplementary Figure 2.

Multiple sequence alignment of the sensor histidine kinases within *Dm*REDusk, *Ac*NIRusk+0, and *Av*NIRusk+0.

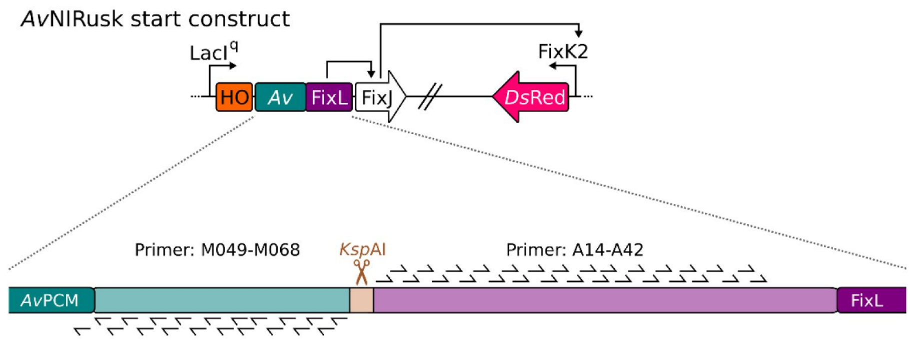
The different shading marks the fusion of the photosensory core modules (PCM) of the bacteriophytochromes from *Deinococcus maricopensis* (*Dm*PCM), *Azorhizobium caulinodans* (*Ac*PCM), and *Agrobacterium vitis* (*Av*PCM) with the catalytic domains of the sensor histidine kinase FixL from *Bradyrhizobium japonicum*. The linker region is shown in pink, and the active-site histidine is marked with a red triangle.

a



Plasmid	Label	501	257	291	Expression
MH013	sc	VLRHN ELLADERGKADVRQRLLNE	TEHQQTQARLQELQSELVHVSRLSAMGEMASALA	HELN	No
MH021	−4	VLRHN ELLADERG	RLSAMGEMASALA	HELN	NIR/Red
MH019	−1a	VLRHN ELLAD	ELVHVSRLSAMGEMASALA	HELN	NIR
MH034	−1b	VLRHN ELL	QSELVHVSRLSAMGEMASALA	HELN	NIR/Red
MH005	0	VLRHN	LQELQSELVHVSRLSAMGEMASALA	HELN	Dark/NIR/Red
MH017	+16a	VLRHN ELLADER	TEHQQTQARLQELQSELVHVSRLSAMGEMASALA	HELN	Dark/Red
MH018	+16b	VLRHN ELLADERG	EHQQTQARLQELQSELVHVSRLSAMGEMASALA	HELN	Dark

b



Plasmid	Label	499	257	291	Expression
MH014	sc	VLRHN ELMQEERNKADIRQRVLNE	TEHQQTQARLQELQSELVHVSRLSAMGEMASALA	HELN	No
MH030	−5	VLRHN ELMQEER	RLSAMGEMASALA	HELN	NIR/Red
MH023	−1	VLRHN ELMQE	ELVHVSRLSAMGEMASALA	HELN	NIR
MH006	0	VLRHN	LQELQSELVHVSRLSAMGEMASALA	HELN	Dark/NIR/Red
MH029	+16a	VLRHN ELMQEERN	EHQQTQARLQELQSELVHVSRLSAMGEMASALA	HELN	Dark/Red
MH020	+16b	VLRHN ELMQEER	TEHQQTQARLQELQSELVHVSRLSAMGEMASALA	HELN	Dark/Red
MH028	+17a	VLRHN ELMQEERK	TEHQQTQARLQELQSELVHVSRLSAMGEMASALA	HELN	NIR
MH038	+17b	VLRHN ELMQEERN	TEHQQTQARLQELQSELVHVSRLSAMGEMASALA	HELN	NIR/Red

Supplementary Figure 3.

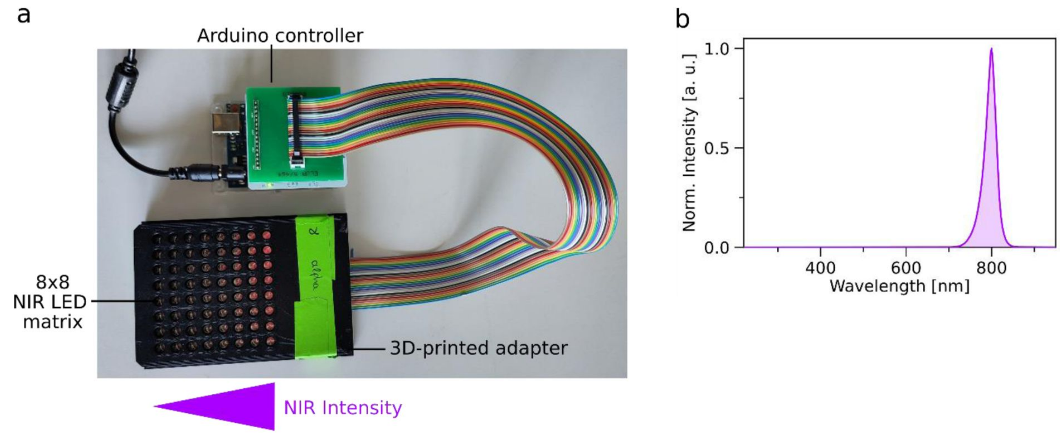
***Ac*NIRusk (a) and *Av*NIRusk (b) PATCHY start constructs (sc) and derived linker variants.**

The linker between the *Ac*/*Av*PCM and FixL was adopted from the parental proteins and is defined as the region between the PCM (residues VLRHN) and the active-site histidine at position 291. At the junction, a *Ksp*AI restriction site and a frameshift were inserted. Underlined amino acids originate from a cloning artifact. PATCHY primers are listed in **Suppl. Table 1**.

Supplementary Figure 4.

Setup for illumination with near-infrared (NIR) light at different intensities in a microtiter-plate (MTP) format.

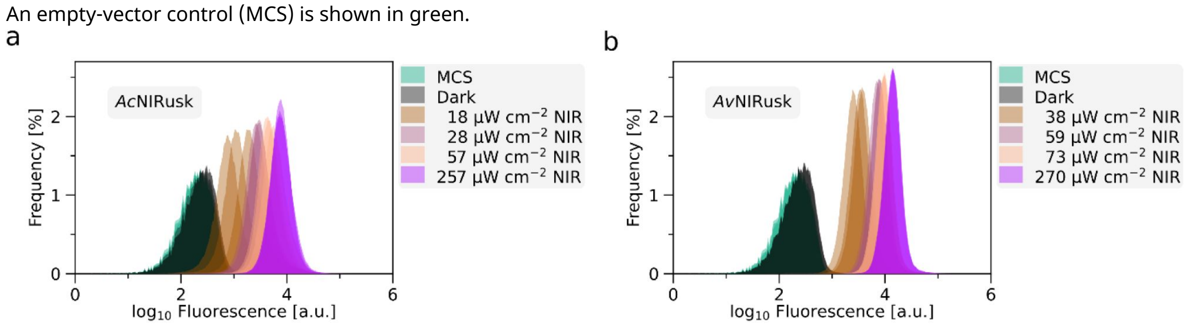
(a) The programmable 8-by-8 matrix of NIR light-emitting diodes (LED) is based on a previous setup (Hennemann et al., 2018; Stüven et al., 2019) and integrated into a 3D-printed adapter. NIR-light intensities can be adjusted by an Arduino microcontroller. (b) Emission spectrum of the NIR LED matrix with a maximum at 800 nm and a full width at half-maximum of 18 nm.



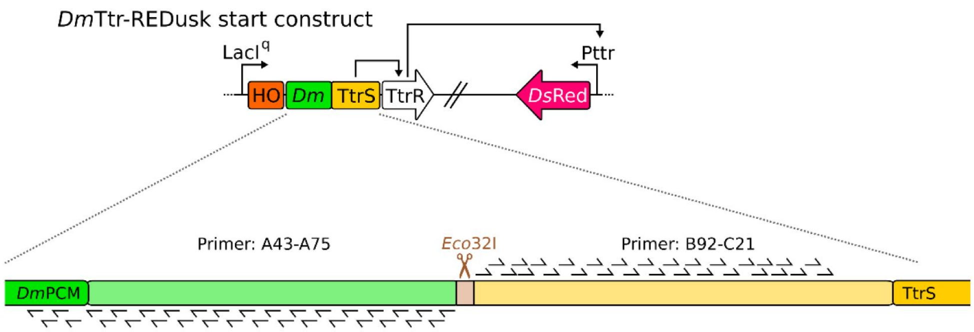
Supplementary Figure 5.

Analyses by flow cytometry of *Ds*Red production of bacteria containing *Ac*NIRusk (a) or *Av*NIRusk (b) when incubated in darkness (black), or under different NIR-light intensities.

An empty-vector control (MCS) is shown in green.

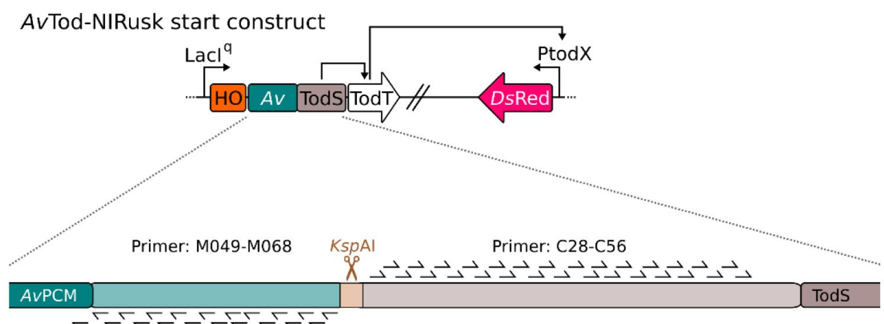


a



Plasmid	Label	510	407	441	Expression
SM123	sc	GARL AALQDLNARLERANRELEGRNAELRRFAF	RELINTQRQLNENRALLEHAQRIAIAGELGASLS	HELN	No
SM146	−13	GARL A	AIAGELGASLS	HELN	Red
SM147	0	GARL	LNENRALLEHAQRIAIAGELGASLS	HELN	Dark/NIR
SM148	+5	GARL AA	QRQLNENRALLEHAQRIAIAGELGASLS	HELN	Red
SM149	+7a	GARL A	INTQRQLNENRALLEHAQRIAIAGELGASLS	HELN	Dark/NIR
SM150	+7b	GARL AALQDLNARLERANRELEGRNAELRR	LGASLS	HELN	Dark/NIR
SM153	+14	GARL AALQDLNARLERANRELEGRNAELRRFAFD	AGELGASLS	HELN	Dark/NIR

b

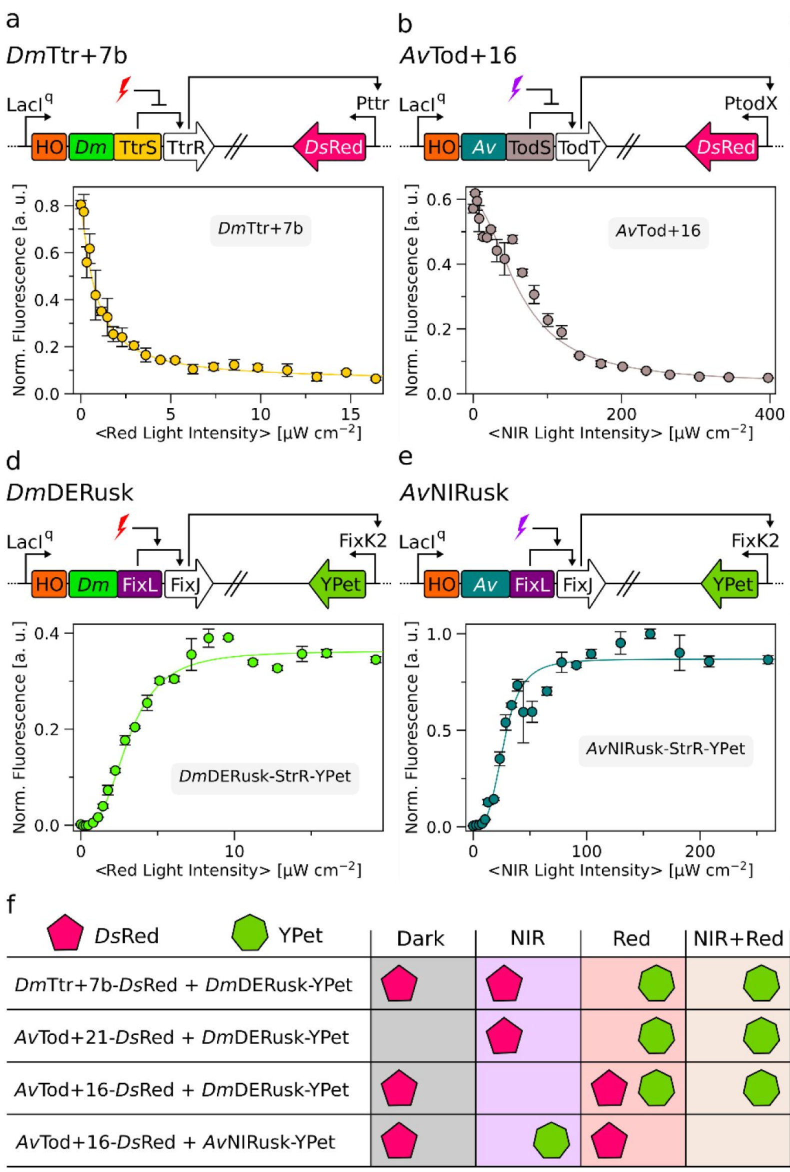


Plasmid	Label	499	726	760	Expression
SM128	sc	VLRHN ELMQEERNKADIRQRVLNE	TEKKQAQENLNQLQQQLVYVSRSATMGEFAAYIA	HEIN	No
SM160	+3	VLRHN ELMQEERNKADIRQRVLNE	MGEFAAYIA	HEIN	NIR
SM161	+16	VLRHN ELMQEERNKADIRQRVLNEV	QQQLVYVSRSATMGEFAAYIA	HEIN	Dark/Red
SM162	+19	VLRHN ELMQEERNKADIRQRVLNEV	NQLQQQLVYVSRSATMGEFAAYIA	HEIN	Dark
SM163	+21	VLRHN ELMQEERNKADIRQR	KQAQENLNQLQQQLVYVSRSATMGEFAAYIA	HEIN	NIR/Red
SM164	+26	VLRHN ELMQEERNKADIRQRVL	TEKKQAQENLNQLQQQLVYVSRSATMGEFAAYIA	HEIN	Dark/Red

Supplementary Figure 6.

***Dm*Ttr-REDusk (a) and *Av*Tod-NIRusk (b) PATCHY start constructs (sc) and derived linker variants.**

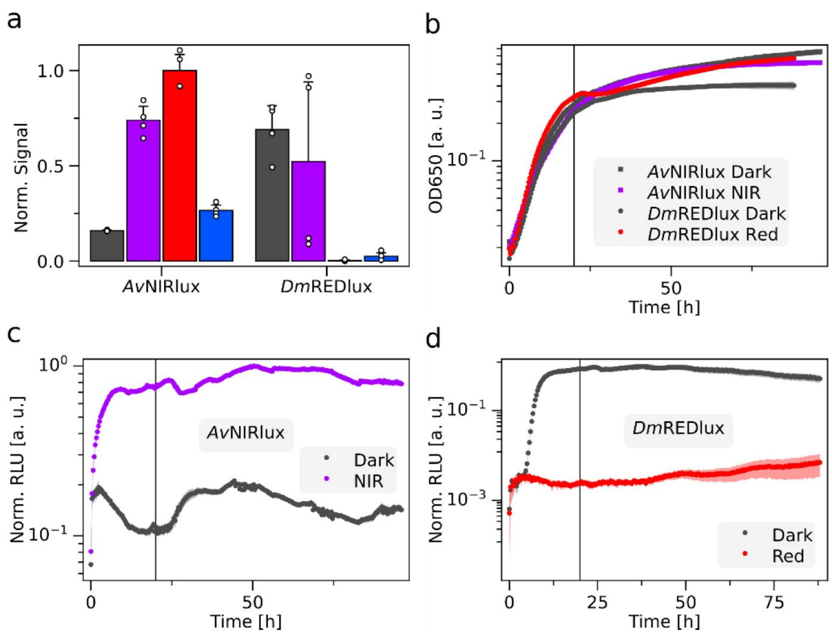
The linker between the *Dm*/*Av*PCM and the TtrS/TodS two-component system was adopted from the natural proteins. At the junction, a *Eco*32I or *Ksp*AI restriction site and a frameshift were inserted. Underlined amino acids highlighted derive from a cloning artifact. PATCHY primers are listed in **Suppl. Table 1**.



Supplementary Figure 7.

Characterization of constructs used in multiplexing experiments.

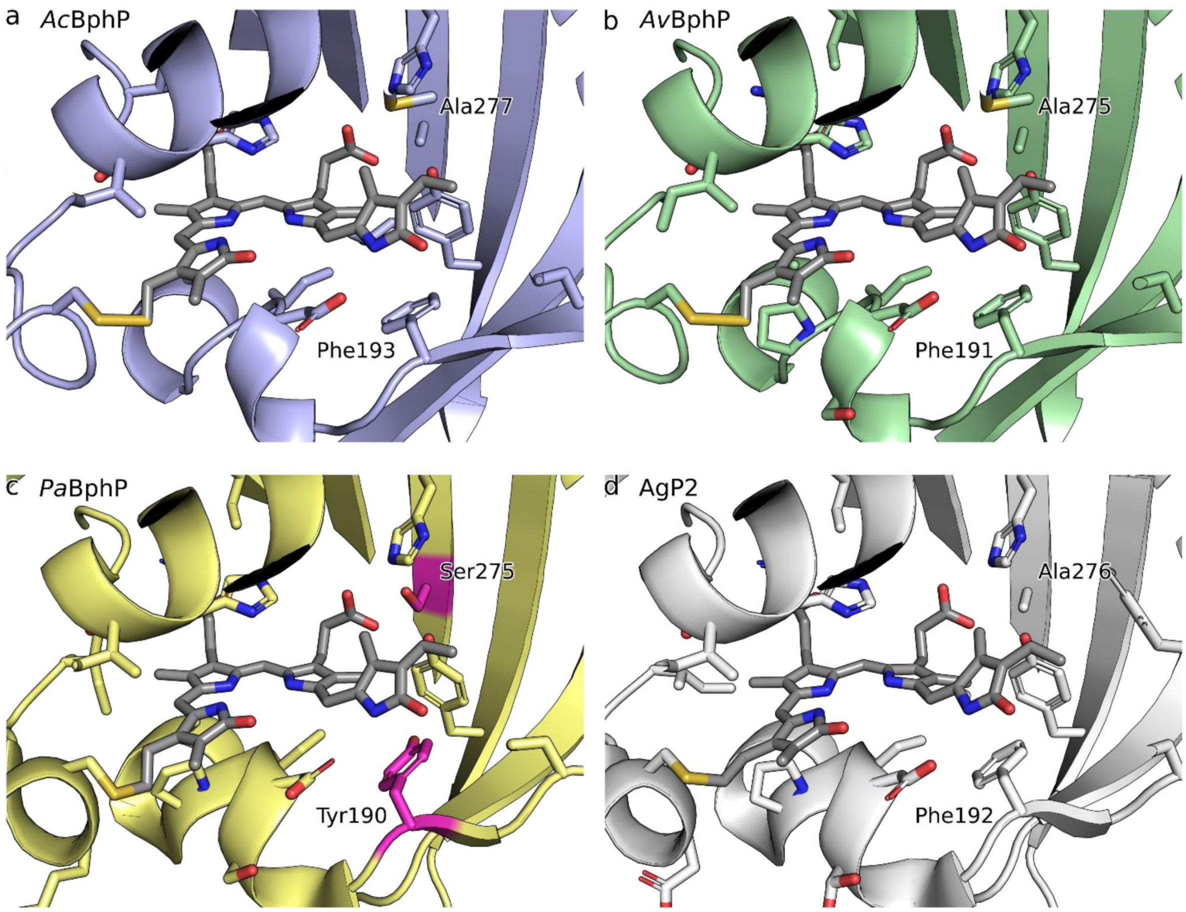
(a) Schematic of *Dm*Ttr+7b (top) and associated light-dose response (bottom). (b) and (c) as in (a) but for *Av*Tod+16 and *Av*Tod+21. Data in panels a-c are normalized to the maximal fluorescence obtained for *Av*Tod+21. (d) Schematic of *Dm*DERusk-StrR-YPet (top) and the associated light-dose response (bottom). (e) as in (d) but for *Av*NIRusk-StrR-YPet. Data in panels d-f are normalized to the maximal fluorescence obtained for of *Av*NIRusk-StrR-YPet. (f) The schematic summarizes the results of the multiplexing experiments from **Fig. 3c**. The production of *Ds*Red and YPet under different light conditions are shown by pink pentagons and green heptagons, respectively.



Supplementary Figure 8.

Gene expression regulated by *Av*NIRlux and *Dm*REDlux in *Agrobacterium tumefaciens*.

(a) Quantified luminescence from the spotting assay in **Fig. 4b**. Bar colors represent incubation in darkness (gray), NIR (purple), red (red), or blue (blue) light. (b) Optical density at 650 nm of *A. tumefaciens* bacteria bearing *Av*NIRlux or *Dm*REDlux grown in liquid culture under different light conditions. (c) Normalized reporter luminescence of the bacteria from panel (b) harboring *Av*NIRlux and grown in darkness (grey) or under NIR light (purple). The vertical black line denotes a time point of 20 h on which the data in **Fig. 4d** are based. (d) as for (c) but for *A. tumefaciens* bacteria containing *Dm*REDlux and illumination with red instead of NIR light (red).



Supplementary Figure 9.

Biliverdin binding pocket in the photosensory core modules (PCM) of (a) *Azorhizobium caulinodans* bacteriophytochrome (BphP), (b) *Agrobacterium vitis* BphP, (c) *Pseudomonas aerugionsa* BphP, and (d) *Agrobacterium tumefaciens* Agp2 BphP.

Whereas experimental structures are available for the PCMs of AgP2 [PDB entry 6g1y (Schmidt et al., 2018)] and *Pa*BphP [3c2w (Yang et al., 2008)], for *Ac*BphP and *Av*BphP, PCM models were calculated with AlphaFold3 (Abramson et al., 2024). Tyrosine 190 and serine 275 are highlighted in the *Pa*PCM structure by magenta color. The other three PCMs feature less polar phenylalanine and alanine residues, respectively, in the structurally equivalent positions.

Name	Sequence 5'-3'
Primers for cloning of bathy-PCM expression constructs	
M001_Ac_Bb_for	CCTGGTTGAAATTGTGCTGCGTCATAATGAGAATCTTTATTTTCAGCATCACCA
M002_Ac_Bb_rev	GATCAACTGCTTCTGCATTCGGCATGGTATATCTCCTTATTAAAGTTAAACAAAA
M003_Ac_Ins_for	TTTTGTTTAACTTTAATAAGGAGATATACCATGCCGAATGCAGAAGCAGTTGATC
M004_Ac_Ins_rev	TGGTGATGCTGAAAATAAAGATTCTCATTATGACGCAGCACAATTTCAACCAGG
M005_Av_Bb_for	ACCGTTAAAGTGGTTCTGCGCCATAATGAGAATCTTTATTTTCAGCATCACCA
M006_Av_Bb_rev	ATCAACGGTCTGGGTTGCAGGCATGGTATATCTCCTTATTAAAGTTAAACAAAA
M007_Av_Ins_for	TTTTGTTTAACTTTAATAAGGAGATATACCATGCCTGCAACCCAGACCGTTGAT
M008_Av_Ins_rev	TGGTGATGCTGAAAATAAAGATTCTCATTATGGCGCAGAACCACTTTAACGGT
M069_Pa-Ins-fw	TTTTGTTTAACTTTAATAAGGAGATATACCATGACGAGCATCACCCCGGTTACCC
M070_Pa-Ins-rev	TGGTGATGGCCCTGAAAATAAAGATTCTCCGCATGGTTGAGGCACAGTTCCATC
M071_Pa-Bb-fw	GATGGAACTGTGCCTCAACCATGCGGAGAATCTTTATTTTCAGGGCCATCACCA
M072_Pa-Bb-rev	GGGTAACCGGGGTGATGCTCGTCATGGTATATCTCCTTATTAAAGTTAAACAAAA
M017_AcAv_TEVrepair_fw	GGCCATCACCATCACCATCACCATCAC
M018_Ac_TEVrepair_rev	CTGAAAATAAAGATTCTCATTATGACGC
M019_Av_TEVrepair_rev	CTGAAAATAAAGATTCTCATTATGGCGC
Primers for Gibson cloning of bathy-PCMs into pREDusk	
M009_Ac_Bb_for	CCTGGTTGAAATTGTGCTGCGTCATAATCTCCAGGAACTGCAATCCGAGCTCGTC
M010_Ac_Bb_rev	TTGTCAGATCAACTGCTTCTGCATTCGTCATGCGTGGGCGACCTCAGGC
M011_Ac_Ins_for	GCCTGAGGTCGCCCACGCATGACGAATGCAGAAGCAGTTGATCTGACAA
M012_Ac_Ins_rev	GACGAGCTCGGATTGCAGTTCCTGGAGATTATGACGCAGCACAATTTCAACCAGG

Supplementary Table 1.

Oligonucleotides used in this study. Overhang regions are indicated in bold, and restriction enzyme recognition sites are underlined.

M013_Av_Bb_for	CCGTTAAAGTGGTTCTGCGCCATAATCTCCAGGAACTGCAATCCGAGCTCGTC
M014_Av_Bb_rev	GTCAGATCAACGGTCTGGGTTGCAGTCATGCGTGGGCGACCTCAGGC
M015_Av_Ins_for	GCCTGAGGTCGCCCACGCATGACTGCAACCCAGACCGTTGATCTGAC
M016_Av_Ins_rev	GACGAGCTCGGATTGCAGTTCCTGGAGATTATGGCGCAGAACCACTTTAACG G
Primers for cloning of PATCHY *Ac*/*Av*NIRusk start constructs	
A07_P_I_II_Step1_for	TTAACCGAGCATCAGCAGACACAAGCACGTCTCCAGGAACTGCAATCCGAGC TCG
M020_AcREDusk_PATCHY_rev1	CGCTCGTCAGCCAGTAATTCATTATGACGCAGCACAATTTCAACC
M021_AcREDusk_PATCHY_rev2	GACGCACGTCTGCTTTCCCGCGCTCGTCAGCCAGTAATTCATTAT
M022_AcREDusk_PATCHY_fw2	AGCGTCTTCTTAATGAAGTTAACCGAGCATCAGCAGACACAAG
M023_AvREDusk_PATCHY_rev1	CGCTCCTCTTGCATCAGTTCATTATGGCGCAGAACCACTTTAACG
M025_AvREDusk_PATCHY_fw2	AGCGTGTCTTAAACGAGGTTAACCGAGCATCAGCAGACACAAG
M028_AvREDusk_PATCHY_rev2	GACGAATGTCTGCTTTGTTACGCTCCTCTTGCATCAGTTCATTAT
PATCHY forward primers for amplification of FixL	
A14_P_I_II_for_1	ACCGAGCATCAGCAGAC
A15_P_I_II_for_2	GAGCATCAGCAGACACAAG
A16_P_I_II_for_3	CATCAGCAGACACAAGC
A17_P_I_II_for_4	CAGCAGACACAAGCACG
A18_P_I_II_for_5	CAGACACAAGCACGTCTC
A19_P_I_II_for_6	ACACAAGCACGTCTCC
A20_P_I_II_for_7	CAAGCACGTCTCCAGG
A21_P_I_II_for_8	GCACGTCTCCAGGAAC
A22_P_I_II_for_9	CGTCTCCAGGAACTGC
A23_P_I_II_for_10	CTCCAGGAACTGCAATCC
A24_P_I_II_for_11	CAGGAACTGCAATCCG
A25_P_I_II_for_12	GAACTGCAATCCGAGC
A26_P_I_II_for_13	CTGCAATCCGAGCTCG
A27_P_I_II_for_14	CAATCCGAGCTCGTCC
A28_P_I_II_for_15	TCCGAGCTCGTCCACG
A29_P_I_II_for_16	GAGCTCGTCCACGTCTC

Supplementary Table 1. (continued)

A30_P_I_II_for_17	CTCGTCCACGTCTCCAG
A31_P_I_II_for_18	GTCCACGTCTCCAGGC
A32_P_I_II_for_19	CACGTCTCCAGGCTGAG
A33_P_I_II_for_20	GTCTCCAGGCTGAGCG
A34_P_I_II_for_21	TCCAGGCTGAGCGCCATG
A35_P_I_II_for_22	AGGCTGAGCGCCATGG
A36_P_I_II_for_23	CTGAGCGCCATGGGCG
A37_P_I_II_for_24	AGCGCCATGGGCGAAATG
A38_P_I_II_for_25	GCCATGGGCGAAATGG
A39_P_I_II_for_26	ATGGGCGAAATGGCGTC
A40_P_I_II_for_27	GGCGAAATGGCGTCCG
A41_P_I_II_for_28	GAAATGGCGTCCGCGC
A42_P_I_II_for_29	ATGGCGTCCGCGCTCG
PATCHY forward primers for amplification of TtrS	
B92_P_Ttr_for_01	CGCGAACTGATTAATACCC
B93_P_Ttr_for_02	GAACTGATTAATACCCAGCG
B94_P_Ttr_for_03	CTGATTAATACCCAGCGTCAG
B95_P_Ttr_for_04	ATTAATACCCAGCGTCAGC
B96_P_Ttr_for_05	AATACCCAGCGTCAGC
B97_P_Ttr_for_06	ACCCAGCGTCAGCTGAATG
B98_P_Ttr_for_07	CAGCGTCAGCTGAATG
B99_P_Ttr_for_08	CGTCAGCTGAATGAAAATCG
C01_P_Ttr_for_09	CAGCTGAATGAAAATCGTG
C02_P_Ttr_for_10	CTGAATGAAAATCGTGCACTG
C03_P_Ttr_for_11	AATGAAAATCGTGCACTGC
C04_P_Ttr_for_12	GAAAATCGTGCACTGCTG
C05_P_Ttr_for_13	AATCGTGCACTGCTGG
C06_P_Ttr_for_14	CGTGCACTGCTGGAAC
C07_P_Ttr_for_15	GCACTGCTGGAACATGC
C08_P_Ttr_for_16	CTGCTGGAACATGCACAG
C09_P_Ttr_for_17	CTGGAACATGCACAGC
C10_P_Ttr_for_18	GAACATGCACAGCGTATTG
C11_P_Ttr_for_19	CATGCACAGCGTATTGC
C12_P_Ttr_for_20	GCACAGCGTATTGCCATTG
C13_P_Ttr_for_21	CAGCGTATTGCCATTGC

Supplementary Table 1. (continued)

C14_P_Ttr_for_22	CGTATTGCCATTGCCG
C15_P_Ttr_for_23	ATTGCCATTGCCGGTG
C16_P_Ttr_for_24	GCCATTGCCGGTGAAC
C17_P_Ttr_for_25	ATTGCCGGTGAACTGG
C18_P_Ttr_for_26	GCCGGTGAACTGGGTG
C19_P_Ttr_for_27	GGTGAACTGGGTGCAAG
C20_P_Ttr_for_28	GAACTGGGTGCAAGCC
C21_P_Ttr_for_29	CTGGGTGCAAGCCTGAG
PATCHY forward primers for amplification of TodS	
C28_P_Tod_for_01	ACTGAGAAGAAACAAGCACAGG
C29_P_Tod_for_02	GAGAAGAAACAAGCACAGG
C30_P_Tod_for_03	AAGAAACAAGCACAGGAAAATC
C31_P_Tod_for_04	AAACAAGCACAGGAAAATCTTAACC
C32_P_Tod_for_05	CAAGCACAGGAAAATCTTAACC
C33_P_Tod_for_06	GCACAGGAAAATCTTAACCAG
C34_P_Tod_for_07	CAGGAAAATCTTAACCAGTTGC
C35_P_Tod_for_08	GAAAATCTTAACCAGTTGCAGC
C36_P_Tod_for_09	AATCTTAACCAGTTGCAGCAAC
C37_P_Tod_for_10	CTTAACCAGTTGCAGCAACAAC
C38_P_Tod_for_11	AACCAGTTGCAGCAACAAC
C39_P_Tod_for_12	CAGTTGCAGCAACAACTTG
C40_P_Tod_for_13	TTGCAGCAACAACTTGTG
C41_P_Tod_for_14	CAGCAACAACTTGTGTACG
C42_P_Tod_for_15	CAACAACTTGTGTACGTTTCC
C43_P_Tod_for_16	CAACTTGTGTACGTTTCCC
C44_P_Tod_for_17	CTTGTGTACGTTTCCCGATC
C45_P_Tod_for_18	GTGTACGTTTCCCGATCAG
C46_P_Tod_for_19	TACGTTTCCCGATCAGC
C47_P_Tod_for_20	GTTTCCCGATCAGCTACG
C48_P_Tod_for_21	TCCCGATCAGCTACGATG
C49_P_Tod_for_22	CGATCAGCTACGATGGG
C50_P_Tod_for_23	TCAGCTACGATGGGTGAATTTG
C51_P_Tod_for_24	GCTACGATGGGTGAATTTG
C52_P_Tod_for_25	ACGATGGGTGAATTTGC
C53_P_Tod_for_26	ATGGGTGAATTTGCAGC

Supplementary Table 1. (continued)

C54_P_Tod_for_27	GGTGAATTTGCAGCCTATATTG
C55_P_Tod_for_28	GAATTTGCAGCCTATATTGC
C56_P_Tod_for_29	TTTGCAGCCTATATTGCACAC
PATCHY reverse primers for amplification of *Ac*PCM	
M029_Ac_rev1	TTCATTAAGAAGACGCTGACG
M030_Ac_rev2	ATTAAGAAGACGCTGACGC
M031_Ac_rev3	AAGAAGACGCTGACGC
M032_Ac_rev4	AAGACGCTGACGCACG
M033_Ac_rev5	ACGCTGACGCACGTCTG
M034_Ac_rev6	CTGACGCACGTCTGCTTTC
M035_Ac_rev7	ACGCACGTCTGCTTTCC
M036_Ac_rev8	CACGTCTGCTTTCCCG
M037_Ac_rev9	GTCTGCTTTCCCGCGC
M038_Ac_rev10	TGCTTTCCCGCGCTCG
M039_Ac_rev11	TTTCCCGCGCTCGTCAG
M040_Ac_rev12	CCCGCGCTCGTCAGCC
M041_Ac_rev13	GCGCTCGTCAGCCAGTAATTC
M042_Ac_rev14	CTCGTCAGCCAGTAATTCATTATG
M043_Ac_rev15	GTCAGCCAGTAATTCATTATGACG
M044_Ac_rev16	AGCCAGTAATTCATTATGACG
M045_Ac_rev17	CAGTAATTCATTATGACGCAGC
M046_Ac_rev18	TAATTCATTATGACGCAGCAC
M047_Ac_rev19	TTCATTATGACGCAGCAC
M048_Ac_rev20	ATTATGACGCAGCACAATTTC
PATCHY reverse primers for amplification of *Av*PCM	
M049_Av_rev1	CTCGTTTAAGACACGCTGAC
M050_Av_rev2	GTTTAAGACACGCTGACGAATG
M051_Av_rev3	TAAGACACGCTGACGAATG
M052_Av_rev4	GACACGCTGACGAATGTC
M053_Av_rev5	ACGCTGACGAATGTCTGC
M054_Av_rev6	CTGACGAATGTCTGCTTTG
M055_Av_rev7	ACGAATGTCTGCTTTGTTACG
M056_Av_rev8	AATGTCTGCTTTGTTACGC
M057_Av_rev9	GTCTGCTTTGTTACGCTCC

Supplementary Table 1. (continued)

M058_Av_rev10	TGCTTTGTTACGCTCCTC
M059_Av_rev11	TTTGTTACGCTCCTCTTGC
M060_Av_rev12	GTTACGCTCCTCTTGCATC
M061_Av_rev13	ACGCTCCTCTTGCATCAG
M062_Av_rev14	CTCCTCTTGCATCAGTTCATTATG
M063_Av_rev15	CTCTTGCATCAGTTCATTATGG
M064_Av_rev16	TTGCATCAGTTCATTATGGC
M065_Av_rev17	CATCAGTTCATTATGGCGC
M066_Av_rev18	CAGTTCATTATGGCGCAG
M067_Av_rev19	TTCATTATGGCGCAGAACC
M068_Av_rev20	ATTATGGCGCAGAACC
Primers for cloning of orthogonal TtrSR TCS	
B55_Pttr_BglII_for	CGTTAAAGATCTATATTTGTTGCGCTAGA
B56_Pttr_XbaI_rev	TGCCAGTCTAGAACAGGCATTC
B57_Bb_for_DmTtrSR	ACGTCTGCGTCTGGAAGATAATCGTTAACAATTGATGTAAGTTAGCTCACTCA
B58_Bb_rev_DmTtrSR	CATTCAGCTGACGCTGGGTATTAATCAGCTCACGATTAGCACGCTCTAAG
B59_Ins_for_DmTtrSR	CTTAGAGCGTGCTAATCGTGAGCTGATTAATACCCAGCGTCAGCTGAATG
B60_Ins_rev_DmTtrSR	TGAGTGAGCTAACTTACATCAATTGTTAACGATTATCTTCCAGACGCAGACGT
B84_P_V_sc_for	GGCGTTTCGCCTTTGATATCGCGAACTGATTAATACCCAGCGTCAG
B85_P_V_sc_rev	GCAACTCTGCGTTACGCCCTTCAAGCTCACGATTAGCACGCTCTA
Primers for cloning of orthogonal TodST TCS	
TodX_fw	GCACAAGATCTGGTCTGAGGTTTTCATCGAC
TodX_rv	CGTGCTCTAGAAATTACAATCCTTCCAC
0_pDusk_fw	GAGTCAATTGAGGGTGGTGAATGTGGCTAG
1_TodS_fw	GATATCACCGAGAAGAAACAAGCACAGGAAAATC
2_NoMfeI_rv	CGTACACAAGTTGTTGCTGCAACTGGTTAAGATTTTCCTGTGCTTGTTTC
3_TodS_fw	GCAGCAACAACTTGTGTACGTTTCCCGATC
4_TodT_rv	GCATCAATTGCTATTCCAGGCTATCCTTGAG
5_pDusk_rv	GCTTGTTTCTTCTCGGTGATATCATTCTGAATTC
M014_Av_Bb_rev	GTCAGATCAACGGTCTGGGTTGCAGTCATGCGTGGGCGACCTCAGGC
M015_Av_Ins_for	GCCTGAGGTCGCCCACGCATGACTGCAACCCAGACCGTTGATCTGAC
C26_Bb_for_P_X_sc	ATTCGTCAGCGTGTCTTAAACGAGGTTAACTGAGAAGAAACAAGCACAG
C27_Ins_rev_P_X_sc	CTGTGCTTGTTTCTTCTCAGTTAACCTCGTTTAAGACACGCTGACGAAT

Supplementary Table 1. (continued)

Primers for cloning StrR variants	
B73_Bb_for_pStrepDusk	GATCACCAAGGTAGTCGGCAAATAAGAATTAATTCATGAGCGGATACATATT
B74_Bb_rev_pStrepDusk	AAGGATCTCAAGAAGATCCTTTGATCTGATGCCTCCGTGTAAGGGG
B75_Ins_for_pStrepDusk	CCCCTTACACGGAGGCATCAGATCAAAGGATCTTCTTGAGATCCTT
B76_Ins_rev_pStrepDusk	AATATGTATCCGCTCATGAATTAATTCTTATTTGCCGACTACCTTGGTGATC
B26_Ins_rev	CGAGATGATAGGAGGTCTAGCATGACGACCAAGGGACATATCTACG
B27_Bb_for	CGTAGATATGTCCCTTGGTCGTCATGCTAGACCTCCTATCATCTCG
M014_Av_Bb_rev	GTCAGATCAACGGTCTGGGTTGCAGTCATGCGTGGGCGACCTCAGGC
M015_Av_Ins_for	GCCTGAGGTCGCCCACGCATGACTGCAACCCAGACCGTTGATCTGAC
Primers for cloning MCS variant	
C76_Bb_for_MCS	ATCAGCAACTACATGAAGGGCTCGCGGCGGCTGCTTGCCG
C77_Ins_rev_MCS	CGGCAAGCAGCCGCCGCGAGCCCTTCATGTAGTTGCTGAT
M014_Av_Bb_rev	GTCAGATCAACGGTCTGGGTTGCAGTCATGCGTGGGCGACCTCAGGC
M015_Av_Ins_for	GCCTGAGGTCGCCCACGCATGACTGCAACCCAGACCGTTGATCTGAC
Primers for cloning lux variants	
lux-op_Nde_for	**GGATCCCATATG** AAATTTGGAAACTTTTTGCTTACATACC
lux-op_Sac_rev	**TGACGAGCTC** TCAACTATCAAACGCTTCG
PfixK2_NdeI_for	**TGACCATATG** ATATCTCCTTCTTAAAGTTAAACAAAATT
PfixK2_rev_OLPmamG	**ACTAAGAGCTAGTAAAGCGAAAAAG** CGGGATCTCGACGCTCTC
DrHo_OLPmamG_for	**CTTTTTCGCTTTACTAGCTCTTAGTTCTCCAATAAATTCCCTGCGAAGCTTAGGAGATCAGTATATG** CTCCGACTCATGATCATGAAGC
FixJ_NotI_rev	**TGACGCGGCCGC** TCAATCGTTGAGCATGCC

Supplementary Table 1. (continued)

ID	Construct	Reference
pBAM2	*p15A* ori, miniTn5, Amp^R, Kan^R	Uebe, unpublished
pBAM2-luxAE	*p15A* ori, miniTn5, Amp^R, Kan^R, *luxABCDE*	(Dziuba et al., 2021)
p*Dm*REDlux	*p15A* ori, miniTn5, Amp^R, Kan^R, PfixK2-*luxABCDE*, PmamDC-*DrhemO-Dmpcm-fixL*	(this study)
p*Av*NIRlux	*p15A* ori, miniTn5, Amp^R, Kan^R, PfixK2-*luxABCDE*, PmamDC-*DrhemO-Avpcm-fixL*	(this study)
pMH005	*Ac*NIRusk-0	(this study)
pMH006	*Av*NIRusk-0	(this study)
pMH008	pCDF-Duet-*Ac*PCM/HO	(this study)
pMH009	pCDF-Duet-*Av*PCM/HO	(this study)
pMH013	*Ac*NIRusk-PATCHY-start construct	(this study)
pMH014	*Av*NIRusk-PATCHY-start construct	(this study)
pMH017	*Ac*NIRusk+16a	(this study)
pMH018	*Ac*NIRusk+16b	(this study)
pMH019	*Ac*NIRusk-1a (*Ac*NIRusk)	(this study)
pMH020	*Av*NIRusk+16b	(this study)
pMH021	*Ac*NIRusk-4	(this study)
pMH023	*Av*NIRusk-1	(this study)
pMH026	pCDF-Duet-*Pa*PCM/HO	(this study)
pMH028	*Av*NIRusk+17a (*Av*NIRusk)	(this study)
pMH029	*Av*NIRusk+16a	(this study)
pMH030	*Av*NIRusk-5	(this study)
pMH034	*Ac*NIRusk-1b	(this study)
pMH038	*Av*NIRusk+17b	(this study)
pMIR66	TodS and TodT proteins	(Ramos-González et al., 2002)
pMIR77	PtodX promoter	(Ramos-González et al., 2002)
pQX037	*Pa*PAC	(Xu et al., 2024)
pQX086	*Av*PAC	(Xu et al., 2024)
pQX091	*Ac*PAC	(Xu et al., 2024)
pRD349	pDusk-YT1-X	(unpublished)

Supplementary Table 2.

Plasmids used in this study.

pSM001	*Dr*REDusk-MCS	(Multamäki et al., 2022)
pSM003	*Dm*REDusk	(Meier et al., 2024a)
pSM024	*Dm*REDusk-MCS	(Meier et al., 2024a)
pSM087	*Dm*DERusk	(Meier et al., 2024a)
pSM090	*Dm*DERusk-YPet	(Meier et al., 2024a)
pSM091	*Dm*DERusk-MCS	(Meier et al., 2024a)
pSM122	*Dm*DERusk-StrR-YPet	(this study)
pSM123	*Dm*Ttr-REDusk-PATCHY-start construct	(this study)
pSM128	*Av*Tod-NIRusk-PATCHY-start construct	(this study)
pSM136	*Av*NIRusk-StrR-YPet	(this study)
pSM146	*Dm*Ttr-13	(this study)
pSM147	*Dm*Ttr-0	(this study)
pSM148	*Dm*Ttr+5	(this study)
pSM149	*Dm*Ttr+7a	(this study)
pSM150	*Dm*Ttr+7b	(this study)
pSM153	*Dm*Ttr+14	(this study)
pSM160	*Av*Tod+3	(this study)
pSM161	*Av*Tod+16	(this study)
pSM162	*Av*Tod+19	(this study)
pSM163	*Av*Tod+21	(this study)
pSM164	*Av*Tod+26	(this study)
pSM182	*Av*NIRusk-MCS	(this study) Addgene: 235084

Supplementary Table 2. (continued)

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Additional information

Author Contributions

S.S.M.M. and A.M. conceived of the project; H.T., R.U., and A.M. acquired funding; S.S.M.M., M.H., C.B., R.U., and A.M. designed research; S.S.M.M., M.H., C.B., E.L.R.D., R.U., and A.M. conducted research; S.S.M.M., M.H., C.B., E.L.R.D., R.U., and A.M. analyzed data; S.S.M.M. and A.M. wrote the manuscript; all authors edited the manuscript.

References

Abramson J, Adler J, Dunger J, Evans R, Green T, Pritzel A, Ronneberger O, Willmore L, Ballard AJ, Bambrick J, Bodenstein SW, Evans DA, Hung C-C, O'Neill M, Reiman D, Tunyasuvunakool K, Wu Z, Žemgulytė A, Arvaniti E, Beattie C, Bertolli O, Bridgland A, Cherepanov A, Congreve M, Cowen-Rivers AI, Cowie A, Figurnov M, Fuchs FB, Gladman H, Jain R, Khan YA, Low CMR, Perlin K, Potapenko A, Savy P, Singh S, Stecula A, Thillaisundaram A, Tong C, Yakneen S, Zhong ED, Zielinski M, Žídek A, Bapst V, Kohli P, Jaderberg M, Hassabis D, Jumper JM (2024) **Accurate structure prediction of biomolecular interactions with AlphaFold 3** *Nature* **630**:493–500 https://doi.org/10.1038/s41586-024-07487-w | Google Scholar

Andersen KR, Leksa NC, Schwartz TU (2013) **Optimized E. coli expression strain LOBSTR eliminates common contaminants from His-tag purification** *Prot Struct Funct Bioinf* **81**:1857–1861 https://doi.org/10.1002/prot.24364 | Google Scholar

Anthamatten D, Hennecke H (1991) **The regulatory status of the fixL- and fixJ-like genes in Bradyrhizobium japonicum may be different from that in Rhizobium meliloti** *Mol Gen Genet* **225**:38–48 Google Scholar

Ash C, Dubec M, Donne K, Bashford T (2017) **Effect of wavelength and beam width on penetration in light-tissue interaction using computational methods** *Lasers Med Sci* **32**:1909–1918 https://doi.org/10.1007/s10103-017-2317-4 | Google Scholar

Berkelman TR, Lagarias JC (1986) **Visualization of bilin-linked peptides and proteins in polyacrylamide gels** *Anal Biochem* **156**:194–201 Google Scholar

Blum M, Andreeva A, Florentino LC, Chuguransky SR, Grego T, Hobbs E, Pinto BL, Orr A, Paysan-Lafosse T, Ponamareva I, Salazar GA, Bordin N, Bork P, Bridge A, Colwell L, Gough J, Haft DH, Letunic I, Llinares-López F, Marchler-Bauer A, Meng-Papaxanthos L, Mi H, Natale DA, Orengo CA, Pandurangan AP, Piovesan D, Rivoire C, Sigrist CJA, Thanki N, Thibaud-Nissen F, Thomas PD, Tosatto SCE, Wu CH, Bateman A (2025) **InterPro: the protein sequence classification resource in 2025** *Nucl Acids Res* **53**:D444–D456 https://doi.org/10.1093/nar/gkae1082 | Google Scholar

Bódizs S, Mészáros P, Grunewald L, Takala H, Westenhoff S (2024) **Cryo-EM structures of a bathy phytochrome histidine kinase reveal a unique light-dependent activation mechanism** *Structure* **32**:1952–1962 https://doi.org/10.1016/j.str.2024.08.008 | Google Scholar

Böhm C, Lehtinen K, Multamäki E, Vanhatalo R, Brander O, Meier SSM, Rumfeldt JA, Möglich A, Takala H. (2025) **Traits and application of bathy phytochromes to bacterial optogenetics** *bioRxiv* https://doi.org/10.1101/2025.05.05.652137 | Google Scholar

Borucki B, von Stetten D, Seibeck S, Lamparter T, Michael N, Mroginski MA, Otto H, Murgida DH, Heyn MP, Hildebrandt P. (2005) **Light-induced proton release of phytochrome is coupled to the transient deprotonation of the tetrapyrrole chromophore** *J Biol Chem* **280**:34358–64 Google Scholar

Burgie ES, Basore K, Rau MJ, Summers B, Mickles AJ, Grigura V, Fitzpatrick JAJ, Vierstra RD (2024) **Signaling by a bacterial phytochrome histidine kinase involves a conformational

cascade reorganizing the dimeric photoreceptor *Nat Commun* **15**:6853 https://doi.org/10.1038/s41467-024-50412-y | Google Scholar

Busch A, Guazzaroni M-E, Lacal J, Ramos JL, Krell T (2009) **The Sensor Kinase TodS Operates by a Multiple Step Phosphorelay Mechanism Involving Two Autokinase Domains*** *J Biol Chem* **284**:10353–10360 https://doi.org/10.1074/jbc.M900521200 | Google Scholar

Buschiazzo A, Trajtenberg F (2019) **Two-Component Sensing and Regulation: How Do Histidine Kinases Talk with Response Regulators at the Molecular Level?** *Annu Rev Micro* **73**:507–528 https://doi.org/10.1146/annurev-micro-091018-054627 | Google Scholar

Chernov KG, Redchuk TA, Omelina ES, Verkhusha VV (2017) **Near-Infrared Fluorescent Proteins, Biosensors, and Optogenetic Tools Engineered from Phytochromes** *Chem Rev* **117**:6423–6446 https://doi.org/10.1021/acs.chemrev.6b00700 | Google Scholar

Cubillos-Ruiz A, Guo T, Sokolovska A, Miller PF, Collins JJ, Lu TK, Lora JM (2021) **Engineering living therapeutics with synthetic biology** *Nat Rev Drug Discov* **20**:941–960 https://doi.org/10.1038/s41573-021-00285-3 | Google Scholar

Cui M, Sun T, Shubin Li, Pan H, Liu J, Zhang X, Li L, Shanshan Li, Wei C, Yu C, Yang C, Ma N, Ma B, Lu S, Chang J, Zhang W, Wang H (2021) **NIR light-responsive bacteria with live bio-glue coatings for precise colonization in the gut** *Cell Rep* **36**:109690 https://doi.org/10.1016/j.celrep.2021.109690 | Google Scholar

Daeffler KN, Galley JD, Sheth RU, Ortiz-Velez LC, Bibb CO, Shroyer NF, Britton RA, Tabor JJ (2017) **Engineering bacterial thiosulfate and tetrathionate sensors for detecting gut inflammation** *Mol Syst Biol* **13**:923 https://doi.org/10.15252/msb.20167416 | Google Scholar

Davis SJ, Vener AV, Vierstra RD (1999) **Bacteriophytochromes: Phytochrome-Like Photoreceptors from Nonphotosynthetic Eubacteria** *Science* **286**:2517–2520 https://doi.org/10.1126/science.286.5449.2517 | Google Scholar

Deisseroth K, Feng G, Majewska AK, Miesenböck G, Ting A, Schnitzer MJ (2006) **Next-generation optical technologies for illuminating genetically targeted brain circuits** *J Neurosci* **26**:10380–10386 https://doi.org/10.1523/JNEUROSCI.3863-06.2006 | Google Scholar

Diensthuber RP, Bommer M, Gleichmann T, Möglich A (2013) **Full-length structure of a sensor histidine kinase pinpoints coaxial coiled coils as signal transducers and modulators** *Structure* **21**:1127–1136 https://doi.org/10.1016/j.str.2013.04.024 | Google Scholar

Dziuba M, Riese CN, Borgert L, Wittchen M, Busche T, Kalinowski J, Uebe R, Schüler D (2021) **The Complex Transcriptional Landscape of Magnetosome Gene Clusters in Magnetospirillum gryphiswaldense** *mSystems* **6**:10–1128 https://doi.org/10.1128/msystems.00893-21 | Google Scholar

Etzl S, Lindner R, Nelson MD, Winkler A (2018) **Structure-guided design and functional characterization of an artificial red light–regulated guanylate/adenylate cyclase for optogenetic applications** *J Biol Chem* **293**:9078–9089 https://doi.org/10.1074/jbc.RA118.003069 | Google Scholar

Gao R, Bouillet S, Stock AM (2019) **Structural Basis of Response Regulator Function** *Annu Rev Micro* **73**:175–197 https://doi.org/10.1146/annurev-micro-020518-115931 | Google Scholar

Gibson DG, Young L, Chuang R-Y, Venter JC, Hutchison CA, Smith HO (2009) **Enzymatic assembly of DNA molecules up to several hundred kilobases** *Nat Methods* **6**:343–345 https://doi.org/10.1038/nmeth.1318 | Google Scholar

Gourinchas G, Etzl S, Göbl C, Vide U, Madl T, Winkler A (2017) **Long-range allosteric signaling in red light-regulated diguanylyl cyclases** *Sci Adv* **3**:e1602498 https://doi.org/10.1126/sciadv.1602498 | Google Scholar

Hamblin MR (2016) **Shining light on the head: Photobiomodulation for brain disorders** *BBA Clinical* **6**:113–124 https://doi.org/10.1016/j.bbacli.2016.09.002 | Google Scholar

Hartsough LA, Park M, Kotlajich MV, Lazar JT, Han B, Lin C-CJ, Musteata E, Gambill L, Wang MC, Tabor JJ (2020) **Optogenetic control of gut bacterial metabolism to promote longevity** *eLife* **9**:e56849 https://doi.org/10.7554/eLife.56849 | Google Scholar

Hennemann J, Iwasaki RS, Grund TN, Diensthuber RP, Richter F, Möglich A (2018) **Optogenetic Control by Pulsed Illumination** *Chembiochem* **19**:1296–1304 https://doi.org/10.1002/cbic.201800030 | Google Scholar

Huber C, Strack M, Schultheiß I, Pielage J, Mechler X, Hornbogen J, Diller R, Frankenberg-Dinkel N (2024) **Darkness inhibits autokinase activity of bacterial bathy phytochromes** *J Biol Chem* **300**:107148 https://doi.org/10.1016/j.jbc.2024.107148 | Google Scholar

Hughes J, Lamparter T, Mittmann F, Hartmann E, Gärtner W, Wilde A, Börner T (1997) **A prokaryotic phytochrome** *Nature* **386**:663 https://doi.org/10.1038/386663a0 | Google Scholar

Hughes J, Winkler A (2024) **New Insight Into Phytochromes: Connecting Structure to Function** *Annu Rev Plant Biol* **75**:153–183 https://doi.org/10.1146/annurev-arplant-070623-110636 | Google Scholar

Jacob-Dubuisson F, Mechaly A, Betton J-M, Antoine R (2018) **Structural insights into the signalling mechanisms of two-component systems** *Nat Rev Micro* **16**:585 https://doi.org/10.1038/s41579-018-0055-7 | Google Scholar

Jayaraman P, Devarajan K, Chua TK, Zhang H, Gunawan E, Poh CL (2016) **Blue light-mediated transcriptional activation and repression of gene expression in bacteria** *Nucl Acids Res* **44**:6994–7005 https://doi.org/10.1093/nar/gkw548 | Google Scholar

Jenal U, Reinders A, Lori C (2017) **Cyclic di-GMP: second messenger extraordinaire** *Nat Rev Microbiol* **15**:271–284 https://doi.org/10.1038/nrmicro.2016.190 | Google Scholar

Karniol B, Vierstra RD (2003) **The pair of bacteriophytochromes from Agrobacterium tumefaciens are histidine kinases with opposing photobiological properties** *Proc Natl Acad Sci U S A* **100**:2807–2812 https://doi.org/10.1073/pnas.0437914100 | Google Scholar

Krell T, Lacal J, Busch A, Silva-Jiménez H, Guazzaroni M-E, Ramos JL (2010) **Bacterial Sensor Kinases: Diversity in the Recognition of Environmental Signals** *Annu Rev Micro* **64**:539–559 https://doi.org/10.1146/annurev.micro.112408.134054 | Google Scholar

Lamparter T, Michael N, Mittmann F, Esteban B (2002) **Phytochrome from Agrobacterium tumefaciens has unusual spectral properties and reveals an N-terminal chromophore**

attachment site *Proc Natl Acad Sci U S A* **99**:11628–11633 https://doi.org/10.1073/pnas.152263999 | Google Scholar

Landry BP, Palanki R, Dyulgyarov N, Hartsough LA, Tabor JJ (2018) **Phosphatase activity tunes two-component system sensor detection threshold** *Nat Commun* **9**:1433 https://doi.org/10.1038/s41467-018-03929-y | Google Scholar

Lau PCK, Wang Y, Patel A, Labbé D, Bergeron H, Brousseau R, Konishi Y, Rawlings M (1997) **A bacterial basic region leucine zipper histidine kinase regulating toluene degradation** *Proc Natl Acad Sci U S A* **94**:1453–1458 https://doi.org/10.1073/pnas.94.4.1453 | Google Scholar

Laub MT, Goulian M (2007) **Specificity in Two-Component Signal Transduction Pathways** *Annu Rev Genet* **41**:121–145 https://doi.org/10.1146/annurev.genet.41.042007.170548 | Google Scholar

Lazar JT, Tabor JJ (2021) **Bacterial two-component systems as sensors for synthetic biology applications** *Curr Opin Syst Biol* **28**:100398 https://doi.org/10.1016/j.coisb.2021.100398 | Google Scholar

Lehtinen K, Nokia MS, Takala H (2022) **Red Light Optogenetics in Neuroscience** *Front Cell Neurosci* **15** Google Scholar

Leopold AV, Chernov KG, Shemetov AA, Verkhusha VV (2019) **Neurotrophin receptor tyrosine kinases regulated with near-infrared light** *Nat Commun* **10**:1–13 https://doi.org/10.1038/s41467-019-08988-3 | Google Scholar

Leopold AV, Thankachan S, Yang C, Gerashchenko D, Verkhusha VV (2022) **A general approach for engineering RTKs optically controlled with far-red light** *Nat Methods* **19**:871–880 https://doi.org/10.1038/s41592-022-01517-z | Google Scholar

Levskaya A, Chevalier AA, Tabor JJ, Simpson ZB, Lavery LA, Levy M, Davidson EA, Scouras A, Ellington AD, Marcotte EM, Voigt CA (2005) **Synthetic biology: Engineering Escherichia coli to see light** *Nature* **438**:441–442 https://doi.org/10.1038/nature04405 | Google Scholar

Liu W, Yuan JS, Stewart Jr CN (2013) **Advanced genetic tools for plant biotechnology** *Nat Rev Genet* **14**:781–793 https://doi.org/10.1038/nrg3583 | Google Scholar

Magaraci MS, Bermudez JG, Yogish D, Pak DH, Mollov V, Tycko J, Issadore D, Mannickarottu SG, Chow BY (2016) **Toolbox for Exploring Modular Gene Regulation in Synthetic Biology Training** *ACS Synth Biol* **5**:781–785 https://doi.org/10.1021/acssynbio.6b00057 | Google Scholar

Malla TN, Hernandez C, Muniyappan S, Menendez D, Bizhga D, Mendez JH, Schwander P, Stojković EA, Schmidt M (2024) **Photoreception and signaling in bacterial phytochrome revealed by single-particle cryo-EM** *Sci Adv* **10**:eadq0653 https://doi.org/10.1126/sciadv.adq0653 | Google Scholar

Meier SSM, Multamäki E, Ranzani AT, Takala H, Möglich A (2024a) **Leveraging the histidine kinase-phosphatase duality to sculpt two-component signaling** *Nat Commun* **15**:4876 https://doi.org/10.1038/s41467-024-49251-8 | Google Scholar

Meier SSM, Multamäki E, Ranzani AT, Takala H, Möglich A (2024b) **Multimodal Control of Bacterial Gene Expression by Red and Blue Light** *Meth Mol Biol* **2760**:463–477 https://doi.org/10.1007/978-1-0716-3658-9_26 | Google Scholar

Möglich A (2019) **Signal transduction in photoreceptor histidine kinases** *Prot Sci* **28**:1923–1946 https://doi.org/10.1002/pro.3705 | Google Scholar

Möglich A (2018) **An Open-Source, Cross-Platform Resource for Nonlinear Least-Squares Curve Fitting** *J Chem Educ* **95**:2273–2278 https://doi.org/10.1021/acs.jchemed.8b00649 | Google Scholar

Möglich A, Ayers RA, Moffat K (2010) **Addition at the molecular level: signal integration in designed Per-ARNT-Sim receptor proteins** *J Mol Biol* **400**:477–486 https://doi.org/10.1016/j.jmb.2010.05.019 | Google Scholar

Möglich A, Ayers RA, Moffat K (2009) **Design and signaling mechanism of light-regulated histidine kinases** *J Mol Biol* **385**:1433–1444 https://doi.org/10.1016/j.jmb.2008.12.017 | Google Scholar

Mukougawa K, Kanamoto H, Kobayashi T, Yokota A, Kohchi T (2006) **Metabolic engineering to produce phytochromes with phytochromobilin, phycocyanobilin, or phycoerythrobilin chromophore in Escherichia coli** *FEBS Lett* **580**:1333–1338 https://doi.org/10.1016/j.febslet.2006.01.051 | Google Scholar

Multamäki E, García de Fuentes A, Sieryi O, Bykov A, Gerken U, Ranzani AT, Köhler J, Meglinski I, Möglich A, Takala H. (2022) **Optogenetic Control of Bacterial Expression by Red Light** *ACS Synth Biol* **11**:3354–3367 https://doi.org/10.1021/acssynbio.2c00259 | Google Scholar

Multamäki E, Nanekar R, Morozov D, Lievonen T, Golonka D, Wahlgren WY, Stucki-Buchli B, Rossi J, Hytönen VP, Westenhoff S, Ihalainen JA, Möglich A, Takala H (2021) **Comparative analysis of two paradigm bacteriophytochromes reveals opposite functionalities in two-component signaling** *Nat Commun* **12**:4394 https://doi.org/10.1038/s41467-021-24676-7 | Google Scholar

Nguyen AW, Daugherty PS (2005) **Evolutionary optimization of fluorescent proteins for intracellular FRET** *Nat Biotechnol* **23**:355–360 https://doi.org/10.1038/nbt1066 | Google Scholar

Ohlendorf R, Möglich A (2022) **Light-regulated gene expression in Bacteria: Fundamentals, advances, and perspectives** *Front Bioeng Biotech* **10**:1029403 Google Scholar

Ohlendorf R, Schumacher CH, Richter F, Möglich A (2016) **Library-Aided Probing of Linker Determinants in Hybrid Photoreceptors** *ACS Synth Biol* **5**:1117–1126 https://doi.org/10.1021/acssynbio.6b00028 | Google Scholar

Ohlendorf R, Vidavski RR, Eldar A, Moffat K, Möglich A (2012) **From dusk till dawn: one-plasmid systems for light-regulated gene expression** *J Mol Biol* **416**:534–542 https://doi.org/10.1016/j.jmb.2012.01.001 | Google Scholar

Ong NT, Olson EJ, Tabor JJ (2018) **Engineering an E. coli Near-Infrared Light Sensor** *ACS Synth Biol* **7**:240–248 https://doi.org/10.1021/acssynbio.7b00289 | Google Scholar

Pan H, Li L, Pang G, Han C, Liu B, Zhang Y, Shen Y, Sun T, Liu J, Chang J, Wang H (2021) **Engineered NIR light-responsive bacteria as anti-tumor agent for targeted and precise**

cancer therapy *Chem Eng J* **426**:130842 https://doi.org/10.1016/j.cej.2021.130842 | Google Scholar

Parkinson JS, Kofoid EC (1992) **Communication Modules in Bacterial Signaling Proteins** *Annu Rev Genet* **26**:71–112 https://doi.org/10.1146/annurev.ge.26.120192.000443 | Google Scholar

Ramos-González M-I, Olson M, Gatenby AA, Mosqueda G, Manzanera M, Campos MJ, Víchez S, Ramos JL (2002) **Cross-Regulation between a Novel Two-Component Signal Transduction System for Catabolism of Toluene in Pseudomonas mendocina and the TodST System from Pseudomonas putida** *J Bact* **184**:7062–7067 https://doi.org/10.1128/jb.184.24.7062-7067.2002 | Google Scholar

Ranzani AT, Buchholz K, Blackholm M, Kopkin H, Möglich A (2024) **Induction of bacterial expression at the mRNA level by light** *Nucl Acids Res* **52**:10017–10028 https://doi.org/10.1093/nar/gkae678 | Google Scholar

Rottwinkel G, Oberpichler I, Lamparter T (2010) **Bathy Phytochromes in Rhizobial Soil Bacteria** *J Bact* **192**:5124–5133 https://doi.org/10.1128/JB.00672-10 | Google Scholar

Rumfeldt JA, Takala H, Liukkonen A, Ihalainen JA (2019) **UV-Vis Spectroscopy Reveals a Correlation Between Y263 and BV Protonation States in Bacteriophytochromes** *Photochem Photobiol* **95**:969–979 https://doi.org/10.1111/php.13095 | Google Scholar

Russo FD, Silhavy TJ (1993) **The essential tension: opposed reactions in bacterial two-component regulatory systems** *Trends Microbiol* **1**:306–310 Google Scholar

Sankaran S, del Campo A. (2019) **Optoregulated Protein Release from an Engineered Living Material** *Adv Biosyst* :1800312 https://doi.org/10.1002/adbi.201800312 | Google Scholar

Schindelin J, Arganda-Carreras I, Frise E, Kaynig V, Longair M, Pietzsch T, Preibisch S, Rueden C, Saalfeld S, Schmid B, Tinevez J-Y, White DJ, Hartenstein V, Eliceiri K, Tomancak P, Cardona A (2012) **Fiji: an open-source platform for biological-image analysis** *Nat Methods* **9**:676–682 https://doi.org/10.1038/nmeth.2019 | Google Scholar

Schmidl SR, Sheth RU, Wu A, Tabor JJ (2014) **Refactoring and Optimization of Light-Switchable Escherichia coli Two-Component Systems** *ACS Synth Biol* **3**:820–831 https://doi.org/10.1021/sb500273n | Google Scholar

Schmidt A, Sauthof L, Szczepek M, Lopez MF, Escobar FV, Qureshi BM, Michael N, Buhrke D, Stevens T, Kwiatkowski D, von Stetten D, Mroginski MA, Krauß N, Lamparter T, Hildebrandt P, Scheerer P. (2018) **Structural snapshot of a bacterial phytochrome in its functional intermediate state** *Nat Commun* **9**:4912 https://doi.org/10.1038/s41467-018-07392-7 | Google Scholar

Silva-Jiménez H, Ramos JL, Krell T (2012) **Construction of a prototype two-component system from the phosphorelay system TodS/TodT** *Protein Eng Des Sel* **25**:159–169 https://doi.org/10.1093/protein/gzs001 | Google Scholar

Stabel R, Stüven B, Hansen JN, Körschen HG, Wachten D, Möglich A (2019) **Revisiting and Redesigning Light-Activated Cyclic-Mononucleotide Phosphodiesterases** *J Mol Biol* **431**:3029–3045 https://doi.org/10.1016/j.jmb.2019.07.011 | Google Scholar

Strack RL, Strongin DE, Bhattacharyya D, Tao W, Berman A, Broxmeyer HE, Keenan RJ, Glick BS (2008) **A noncytotoxic DsRed variant for whole-cell labeling** *Nat Methods* **5**:955–957 https://

doi.org/10.1038/nmeth.1264 | Google Scholar

Stüven B, Stabel R, Ohlendorf R, Beck J, Schubert R, Möglich A (2019) **Characterization and engineering of photoactivated adenylyl cyclases** *Biol Chem* **400**:429–441 https://doi.org/10.1515/hsz-2018-0375 | Google Scholar

Tabor JJ, Levskaya A, Voigt CA (2011) **Multichromatic control of gene expression in Escherichia coli** *J Mol Biol* **405**:315–324 https://doi.org/10.1016/j.jmb.2010.10.038 | Google Scholar

Takala H, Edlund P, Ihalainen JA, Westenhoff S (2020) **Tips and turns of bacteriophytochrome photoactivation** *Photochem Photobiol Sci* **19**:1488–1510 https://doi.org/10.1039/D0PP00117A | Google Scholar

Tang K, Beyer HM, Zurbriggen MD, Gärtner W (2021) **The Red Edge: Bilin-Binding Photoreceptors as Optogenetic Tools and Fluorescence Reporters** *Chem Rev* **121**:14906–14956 https://doi.org/10.1021/acs.chemrev.1c00194 | Google Scholar

Tasler R, Moises T, Frankenberg-Dinkel N (2005) **Biochemical and spectroscopic characterization of the bacterial phytochrome of Pseudomonas aeruginosa** *Febs J* **272**:1927–1936 https://doi.org/10.1111/j.1742-4658.2005.04623.x | Google Scholar

Velazquez Escobar F, Piwowarski P, Salewski J, Michael N, Fernandez Lopez M, Rupp A, Qureshi BM, Scheerer P, Bartl F, Frankenberg-Dinkel N, Siebert F, Andrea Mroginski M, Hildebrandt P (2015) **A protonation-coupled feedback mechanism controls the signalling process in bathy phytochromes** *Nature Chem* **7**:423–430 https://doi.org/10.1038/nchem.2225 | Google Scholar

Weissleder R (2001) **A clearer vision for in vivo imaging** *Nat Biotech* **19**:316–317 https://doi.org/10.1038/86684 | Google Scholar

Xu Q, Vogt A, Frechen F, Yi C, Küçükerden M, Ngum N, Sitjà-Roqueta L, Greiner A, Parri R, Masana M, Wenger N, Wachten D, Möglich A (2024) **Engineering Bacteriophytochrome-coupled Photoactivated Adenylyl Cyclases for Enhanced Optogenetic cAMP Modulation. J Mol Biol** *Photosensory Receptors-Mechanisms and Effects* **436**:168257 https://doi.org/10.1016/j.jmb.2023.168257 | Google Scholar

Yang C, Cui M, Zhang Y, Pan H, Liu J, Wang S, Ma N, Chang J, Sun T, Wang H (2020) **Upconversion optogenetic micro-nanosystem optically controls the secretion of light-responsive bacteria for systemic immunity regulation** *Commun Biol* **3**:1–14 https://doi.org/10.1038/s42003-020-01287-4 | Google Scholar

Yang X, Kuk J, Moffat K (2008) **Crystal structure of Pseudomonas aeruginosa bacteriophytochrome: Photoconversion and signal transduction** *Proc Natl Acad Sci U S A* **105**:14715–14720 Google Scholar

Yi C, Meier SSM, Kehr M, Möglich A. (2025) **Spectroscopic Analyses of Photoconversion in Phytochromes** *Zenodo* https://doi.org/10.5281/zenodo.15180372 | Google Scholar

Ziegler T, Möglich A (2015) **Photoreceptor engineering** *Front Mol Biosci* **2**:30 https://doi.org/10.3389/fmolb.2015.00030 | Google Scholar

Zienicke B, Molina I, Glenz R, Singer P, Ehmer D, Escobar FV, Hildebrandt P, Diller R, Lamparter T (2013) **Unusual Spectral Properties of Bacteriophytochrome Agp2 Result from a**

Deprotonation of the Chromophore in the Red-absorbing Form Pr* *J Biol Chem* **288**:31738–31751 https://doi.org/10.1074/jbc.M113.479535 | Google Scholar

Abramson J, Adler J, Dunger J, Evans R, Green T, Pritzel A, Ronneberger O, Willmore L, Ballard AJ, Bambrick J, Bodenstein SW, Evans DA, Hung C-C, O'Neill M, Reiman D, Tunyasuvunakool K, Wu Z, Žemgulytė A, Arvaniti E, Beattie C, Bertolli O, Bridgland A, Cherepanov A, Congreve M, Cowen-Rivers AI, Cowie A, Figurnov M, Fuchs FB, Gladman H, Jain R, Khan YA, Low CMR, Perlin K, Potapenko A, Savy P, Singh S, Stecula A, Thillaisundaram A, Tong C, Yakneen S, Zhong ED, Zielinski M, Žídek A, Bapst V, Kohli P, Jaderberg M, Hassabis D, Jumper JM. (2024) **Accurate structure prediction of biomolecular interactions with AlphaFold 3** *Nature* **630**:493–500 https://doi.org/10.1038/s41586-024-07487-w | Google Scholar

Dziuba M, Riese CN, Borgert L, Wittchen M, Busche T, Kalinowski J, Uebe R, Schüler D (2021) **The Complex Transcriptional Landscape of Magnetosome Gene Clusters in Magnetospirillum gryphiswaldense** *mSystems* **6**:10 https://doi.org/10.1128/msystems.00893-21 | Google Scholar

Hennemann J, Iwasaki RS, Grund TN, Diensthuber RP, Richter F, Möglich A (2018) **Optogenetic Control by Pulsed Illumination** *Chembiochem* **19**:1296–1304 https://doi.org/10.1002/cbic.201800030 | Google Scholar

Multamäki E, de Fuentes A García, Sieryi O, Bykov A, Gerken U, Ranzani AT, Köhler J, Meglinski I, Möglich A, Takala H. (2022) **Optogenetic Control of Bacterial Expression by Red Light** *ACS Synth Biol* **11**:3354–3367 https://doi.org/10.1021/acssynbio.2c00259 | Google Scholar

Meier SSM, Multamäki E, Ranzani AT, Takala H, Möglich A (2024a) **Leveraging the histidine kinase-phosphatase duality to sculpt two-component signaling** *Nat Commun* **15**:4876 https://doi.org/10.1038/s41467-024-49251-8 | Google Scholar

Ramos-González M-I, Olson M, Gatenby AA, Mosqueda G, Manzanera M, Campos MJ, Víchez S, Ramos JL (2002) **Cross-Regulation between a Novel Two-Component Signal Transduction System for Catabolism of Toluene in Pseudomonas mendocina and the TodST System from Pseudomonas putida** *J Bact* **184**:7062–7067 https://doi.org/10.1128/jb.184.24.7062-7067.2002 | Google Scholar

Schmidt A, Sauthof L, Szczepek M, Lopez MF, Escobar FV, Qureshi BM, Michael N, Buhrke D, Stevens T, Kwiatkowski D, von Stetten D, Mroginski MA, Krauß N, Lamparter T, Hildebrandt P, Scheerer P. (2018) **Structural snapshot of a bacterial phytochrome in its functional intermediate state** *Nat Commun* **9**:4912 https://doi.org/10.1038/s41467-018-07392-7 | Google Scholar

Stüven B, Stabel R, Ohlendorf R, Beck J, Schubert R, Möglich A (2019) **Characterization and engineering of photoactivated adenylyl cyclases** *Biol Chem* **400**:429–441 https://doi.org/10.1515/hsz-2018-0375 | Google Scholar

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Yang X, Kuk J, Moffat K (2008) **Crystal structure of Pseudomonas aeruginosa bacteriophytochrome: Photoconversion and signal transduction** *Proc Natl Acad Sci U S A* **105**:14715–14720 Google Scholar

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Reviewer #1 (Public review):

Summary:

This is an interesting study characterizing and engineering so-called bathy phytochromes, i.e., those that respond to near infrared (NIR) light in the ground state, for optogenetic control of bacterial gene expression. Previously, the authors have developed a structure-guided approach to functionally link several light-responsive protein domains to the signaling domain of the histidine kinase FixL, which ultimately controls gene expression. Here, the authors use the same strategy to link bathy phytochrome light-responsive domains to FixL, resulting in sensors of NIR light. Interestingly, they also link these bathy phytochrome light-sensing domains to signaling domains from the tetrathionate-sensing SHK TtrS and the toluene-sensing SHK TodS, demonstrating the generality of their protein engineering approach more broadly across bacterial two-component systems.

This is an exciting result that should inspire future bacterial sensor design. They go on to leverage this result to develop what is, to my knowledge, the first system for orthogonally controlling the expression of two separate genes in the same cell with NIR and Red light, a valuable contribution to the field.

Finally, the authors reveal new details of the pH-dependent photocycle of bathy phytochromes and demonstrate that their sensors work in the gut - and plant-relevant strains E. coli Nissle 1917 and A. tumefaciens.

Strengths:

(1) The experiments are well-founded, well-executed, and rigorous.

(2) The manuscript is clearly written.

(3) The sensors developed exhibit large responses to light, making them valuable tools for ontogenetic applications.

(4) This study is a valuable contribution to photobiology and optogenetics.

Weaknesses:

(1) As the authors note, the sensors are relatively insensitive to NIR light due to the rapid dark reversion process in bathy phytochromes. Though NIR light is generally non-phototoxic, one would expect this characteristic to be a limitation in some downstream applications where light intensities are not high (e.g., in vivo).

(2) Though they can be multiplexed with Red light sensors, these bathy phytochrome NIR sensors are more difficult to multiplex with other commonly used light sensors (e.g., blue) due to the broad light responsivity of the Pfr state. This challenge may be overcome by careful dosing of blue light, as the authors discuss, but other bacterial NIR sensing systems with less cross-talk may be preferred in some applications.

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Reviewer #2 (Public review):

Summary:

In this manuscript, Meier et al. engineer a new class of light-regulated two-component systems. These systems are built using bathy-bacteriophytochromes that respond to near-infrared (NIR) light. Through a combination of genetic engineering and systematic linker optimization, the authors generate bacterial strains capable of selective and tunable gene expression in response to NIR stimulation. Overall, these results are an interesting expansion of the optogenetic toolkit into the NIR range. The cross-species functionality of the system, modularity, and orthogonality have the potential to make these tools useful for a range of applications.

Strengths:

(1) The authors introduce a novel class of near-infrared light-responsive two-component systems in bacteria, expanding the optogenetic toolbox into this spectral range.

(2) Through engineering and linker optimization, the authors achieve specific and tunable gene expression, with minimal cross-activation from red light in some cases.

(3) The authors show that the engineered systems function robustly in multiple bacterial strains, including laboratory E. coli, the probiotic E. coli Nissle 1917, and Agrobacterium tumefaciens.

(4) The combination of orthogonal two-component systems can allow for simultaneous and independent control of multiple gene expression pathways using different wavelengths of light.

(5) The authors explore the photophysical properties of the photosensors, investigating how environmental factors such as pH influence light sensitivity.

Weaknesses:

(1) The expression of multi-gene operons and fluorescent reporters could impose a metabolic burden. The authors should present data comparing optical density for growth curves of engineered strains versus the corresponding empty-vector control to provide insight into the burden and overall impact of the system on host viability and growth.

(2) The manuscript consistently presents normalized fluorescence values, but the method of normalization is not clear (Figure 2 caption describes normalizing to the maximal fluorescence, but the maximum fluorescence of what?). The authors should provide a more detailed explanation of how the raw fluorescence data were processed. In addition, or potentially in exchange for the current presentation, the authors should include the raw fluorescence values in supplementary materials to help readers assess the actual magnitude of the reported responses.

(3) Related to the prior point, it would be useful to have a positive control for fluorescence that could be used to compare results across different figure panels.

(4) Real-time gene expression data are not presented in the current manuscript, but it would be helpful to include a time-course for some of the key designs to help readers assess the speed of response to NIR light.

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Reviewer #3 (Public review):

Summary:

This paper by Meier et al introduces a new optogenetic module for the regulation of bacterial gene expression based on "bathy-BphP" proteins. Their paper begins with a careful characterization of kinetics and pH dependence of a few family members, followed by extensive engineering to produce infrared-regulated transcriptional systems based on the authors' previous design of the pDusk and pDERusk systems, and closing with characterization of the systems in bacterial species relevant for biotechnology.

Strengths:

The paper is important from the perspective of fundamental protein characterization, since bathy-BphPs are relatively poorly characterized compared to their phytochrome and cyanobacteriochrome cousins. It is also important from a technology development perspective: the optogenetic toolbox currently lacks infrared-stimulated transcriptional systems. Infrared light offers two major advantages: it can be multiplexed with additional tools, and it can penetrate into deep tissues with ease relative to the more widely used blue light-activated systems. The experiments are performed carefully, and the manuscript is well written.

Weaknesses:

My major criticism is that some information is difficult to obtain, and some data is presented with limited interpretation, making it difficult to obtain intuition for why certain responses are observed. For example, the changes in red/infrared responses across different figures and cellular contexts are reported but not rationalized. Extensive experiments with variable linker sequences were performed, but the rationale for linker choices was not clearly explained. These are minor weaknesses in an overall very strong paper.

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